The Fertility, Marriage, and Gender Equality Quandary

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Abstract

This paper studies the nexus among fertility, family structure, and gender income equality. I document a novel three-way trade-off: simultaneously achieving high fertility, low single parenthood, and gender income equality is rare across countries and time. Mutual information analysis reveals synergy, meaning the three outcomes should be analyzed jointly. I develop a unified theory explaining this trade-off. Traditional policies—promoting marriage, gender equality, or reducing childcare costs—cannot resolve the tension. However, redistributing childcare from wives to husbands can improve all three outcomes under certain conditions. Calibrating to Mexico (1990–2015), I disentangle gender-neutral from gender-biased technological change.

JEL classification: D13, J11, J12, J13, J16
Keywords: Gender equality, family structure, fertility

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1 Introduction

In the past century, three profound trends underpin the grand gender convergence (Goldin, 2014) observed in many countries: fertility rates have fallen substantially, marriage and stable dual-parent families have become less common, and gender gaps in education and income have steadily narrowed.

Although each of these phenomena—fertility, family structure, and gender income equality—has been studied extensively in isolation or in pairwise combinations, their mutual interdependence has received far less attention. This paper addresses that gap by documenting a fundamental three-way trade-off, developing a unified framework that explains their joint evolution, and—most importantly—identifying conditions under which certain policies can overcome the trade-off.

I begin with empirical analysis. Using large, harmonized datasets covering OECD countries over four decades, I document that very few countries at any time achieve high fertility, widespread dual-parent families, and strong gender income equality simultaneously. The overlap is only a small fraction of what random chance would predict, rejecting the random benchmark following the dartboard approach (Ellison and Glaeser, 1997). Mutual information analysis reveals synergy, meaning the three outcomes should be analyzed jointly rather than in pairs.

To explain these findings, I develop a model in which men and women meet in a marriage market. Building on Becker (1973), men choose between remaining single (and childless) and entering marriage, where they allocate a portion of their income to wives. Women may opt for single motherhood or raise children within marriage. Following Doepke and Kindermann (2019), fertility decisions within marriage are subject to wives' veto. The model connects marriage, fertility, and gender income gaps through two mechanisms: first, within-family transfers and marital fertility jointly clear the marriage market; second, women shoulder the bulk of childcare, so higher fertility reduces female labor supply and widens the gender income gap.

The equilibrium conditions reveal a fundamental tension: there exists a trade-off among high fertility, prevalent dual parenthood, and gender income equality. Pro-marriage policies boost marriage and fertility but widen gender gaps. Gender equality measures narrow gaps but reduce both marriage and fertility. Policies that ease women's childcare burden can mitigate the tension, but their fiscal cost rises disproportionately as wages grow.

The paper's central contribution concerns the design of childcare policies. Traditional approaches—such as subsidizing childcare for women—cannot fully resolve the trade-off because they increase fertility and marriage while widening gender gaps. I show, however, that *redistributing* childcare responsibilities from wives to husbands can potentially achieve all three objectives simultaneously.

I extend the model to allow both spouses to share childcare within marriage. Redistributing childcare to husbands improves all three outcomes under two conditions. First, wives' own labor earnings must exceed the income transfers they receive from husbands—a condition that holds when women have substantial earning power relative to spousal transfers. Second, the husband's utility gain from higher fertility must outweigh his consumption loss from increased childcare time. When both conditions hold, redistributing childcare lowers the wife's effective cost of children, raises marital fertility, increases both spouses' willingness to marry, and narrows gender income gaps as wives work more. Notably, the wife's benefit holds for all values of the elasticity of substitution between consumption and children (provided $\rho > 1$), making the policy prescription robust across a wide range of preference specifications.

This finding has important policy implications. It suggests that the persistent three-way trade-off may be partially an artifact of the gendered division of childcare labor rather than a fundamental economic constraint. Policies promoting paternal involvement—such as paternity leave mandates with "use-it-or-lose-it" provisions—may help societies escape the trade-off, provided the underlying economic conditions are favorable.

However, I also show that redistributing childcare is not universally superior to uniform cost reductions. When wives' earnings are low relative to spousal transfers, the condition for redistribution to benefit wives fails, and traditional subsidies may be more effective. The relative merits depend on the economic environment, cautioning against one-size-fits-all prescriptions.

To illustrate the quantitative implications, I calibrate the model to Mexico's experience from 1990 to 2015—a period of rapid income growth, plunging fertility, rising single motherhood, and a complete reversal of the gender education gap. The equilibrium transfer $\alpha \approx 0.10$ satisfies the condition for wives to benefit throughout the sample, and the calibrated parameters also satisfy the condition for husbands to benefit, suggesting that redistribution policies would be Pareto-improving in this setting. Decomposition reveals that gender-neutral productivity growth explains half of the fertility and dual-parenthood decline, while convergence in education and declining labor market bias drive gender income convergence.

Lastly, I examine intergenerational dynamics by incorporating evidence that boys from single-parent families fare worse than girls in human capital accumulation. This creates a self-reinforcing feedback: declining marriage rates reduce future male human capital, further eroding marriage and fertility. The dynamic model reveals how technological progress can have persistent, amplifying effects across generations.

In sum, this paper delivers a unified view of three defining trends of our time and identifies a promising policy direction. While fertility, family structure, and gender equality are tightly linked through women's time, men's incentives, and marriage market equilibrium, appropriately designed policies that promote more equal sharing of childcare may allow societies to move toward achieving all three objectives simultaneously—though success depends on the underlying economic environment.

2 Motivating Facts

This section documents the empirical relationship among fertility, family structure, and gender income gaps. I employ two complementary statistical approaches—Venn diagrams and mutual information analysis—to establish a fundamental three-way trade-off among these outcomes across multiple samples and measures.

2.1 Data and Methods

Data Sources and Samples

I analyze four complementary samples that vary in geographic scope, time coverage, and variable measurement. Table 1 summarizes the key characteristics of each sample.

Sample	N	Period	Family Structure	Gender Equality
OECD	721	1970–2014	Out-of-wedlock births	Median earnings gap
World	1,130	1990–2015	Children with both parents	Female income share
U.S. States	51	2023	Single-mother households	Median earnings gap
U.S. CZ	741	2000	Single-mother households	Earnings ratio

Table 1. Summary of Data Samples

Notes: CZ = Commuting Zones. All samples use total fertility rate for fertility. OECD data from OECD Family Database. World sample combines UN fertility data, children's living arrangements from Brenøe and Wasserman (2025), and female labor income share from the World Inequality Database (Chancel et al., 2022). U.S. state data from CDC Natality files and American Community Survey. U.S. CZ data from Chetty et al. (2014).

The *OECD sample* comprises an unbalanced panel of 37 countries from 1970 to 2014, yielding 721 country-year observations. For each observation, I collect the total fertility rate, the share of children born out of wedlock (as a proxy for family instability), and the gender earnings gap (the difference between men's and women's median earnings divided by men's median earnings). I apply linear interpolation for missing data points only when the series contained observed values on both sides of the gap. Figure A.1 in the Appendix displays time-series trends in these variables for selected OECD countries, documenting the widespread decline in fertility below replacement level, the sharp increase in out-of-wedlock births, and the steady narrowing of gender earnings gaps.

The *World sample* addresses potential measurement limitations in the OECD data. Out-of-wedlock births do not necessarily imply absent fathers, as children may live in stable cohabiting unions. Moreover, the gender gap in median earnings among workers excludes differences in labor force participation, potentially understating disparities in total labor income. I therefore construct a global panel using direct measures of children's living arrangements from Brenøe and Wasserman (2025), who harmonize census data from IPUMS International covering 95 countries from 1990 to 2019. For gender equality, I use the female labor income share from the World Inequality Database (Chancel et al., 2022), which measures the proportion of aggregate labor income accruing to women. Merging these sources with United Nations fertility data yields a panel of 1,130 country-year observations.

The *U.S. samples* examine whether the trade-off persists in a setting with more homogeneous institutions, culture, and economy. For each of the 50 states plus the District of Columbia, I construct measures using 2023 data: fertility (live births per 1,000 women aged 15–44 from CDC Natality files), family structure (share of children living with a single mother from the American Community Survey), and gender equality (gender gap in median earnings among individuals aged 16 and over). I also analyze 741 commuting zones using year 2000 data to examine finer geographic variation; these results are reported in Appendix A.5.

Variable Construction

Throughout the empirical analysis, I construct binary indicators for each of the three outcomes using sample-specific median cutoffs. This median-based approach ensures balanced categories within each sample and avoids arbitrary absolute thresholds that could bias the analysis. In each sample, I define "high fertility" (F) as observations with fertility above the sample median, "dual parenthood" (M) as observations with family stability above the sample median (or single parenthood below the median), and "gender equality" (G) as observations with gender gaps below the sample median (or female income shares above the median). Table 2 reports the specific cutoff values for each sample.

Statistical Methods

I employ two complementary statistical tools to characterize the joint distribution of the three outcomes.

The first tool is the *Venn diagram*, which displays the joint distribution of the three binary outcomes. Under statistical independence, the probability of achieving all three outcomes

Sample	High Fertility (F)	Dual Parenthood (M)	Gender Equality (G)
OECD	TFR $>$ 1.69	OOW $<$ 31.4%	Gap $<$ 17.2%
World	TFR $>$ 2.81	Both parents $>$ 75.9%	Female share $>$ 32.3%
U.S. States	Births/1000 $>$ 56.9	Single mom $<$ 17%	Gap $<$ 25.6%
U.S. CZ	CBR $>$ 12.9	Single mom $<$ 19.7%	Ratio $<$ 1.80

Table 2. Median Cutoffs for Binary Indicators

Notes: TFR = total fertility rate (children per woman). OOW = out-of-wedlock birth share. CBR = crude birth rate (births per 1,000 population). Gap = (male median earnings − female median earnings) / male median earnings. Ratio = male median earnings / female median earnings.

simultaneously would be $0.5 \times 0.5 \times 0.5 = 12.5\%$—the random benchmark in the dartboard approach (Ellison and Glaeser, 1997). A trinity share substantially below 12.5% indicates a systematic trade-off among the three objectives. I test this formally using bootstrap inference with 5,000 replications.

The second tool is *mutual information analysis* from information theory. While Venn diagrams reveal the scarcity of joint attainment, they do not characterize the *dependency structure* among the three variables. Mutual information quantifies the amount of information that one random variable contains about another, measured in bits. For two binary variables X and Y, the mutual information $I(X;Y)$ equals zero if and only if the variables are statistically independent; higher values indicate stronger dependence.

For three variables, I compute the *interaction information*, which measures how the dependence between any two variables changes when conditioning on the third. Formally, interaction information is defined as the difference between pairwise mutual information and conditional mutual information: $II = I(X;Y) - I(X;Y|Z)$. This quantity can be positive, negative, or zero, with each case having a distinct interpretation. When interaction information is *negative*, the variables exhibit *synergy*: knowing the third variable *increases* the dependence between the other two. Intuitively, synergy means that the three variables together contain information that cannot be decomposed into pairwise relationships. Conversely, when interaction information is *positive*, the variables exhibit *redundancy*: the third variable provides overlapping information that is already captured in the pairwise relationships.

A finding of synergy is particularly important for the three-way trade-off documented in this paper. If the three outcomes exhibit synergy, then analyzing them in pairs—as most prior literature has done—misses crucial information about their joint behavior. The tension among fertility, family structure, and gender equality would be fundamentally a three-way phenomenon that cannot be fully understood from pairwise analysis alone.

2.2 Results

Venn Diagram Analysis

Figure 1 presents Venn diagrams for the three main samples. A consistent pattern emerges: the share of observations achieving all three outcomes (F+M+G) is far below the 12.5% expected under statistical independence.

In the OECD sample, only 2.5% of observations achieve the trinity—less than one-quarter of the independence benchmark. Bootstrap inference yields a z-statistic of -16.51 ($p < 0.001$), decisively rejecting the null of independence. The World sample shows a similarly stark pattern: only 4.3% of observations achieve all three outcomes ($z = -12.40$, $p < 0.001$). The U.S. state-level analysis confirms the trade-off even within a single country: only three states (5.9%) achieve all three outcomes ($z = -1.90$, $p = 0.029$).

Importantly, when testing whether any *pairwise* combination falls below its independence benchmark of 25%, none of the three one-sided tests reject at conventional significance levels in any sample. This indicates that the tension emerges distinctly only when all three outcomes are considered simultaneously.

Mutual Information Analysis

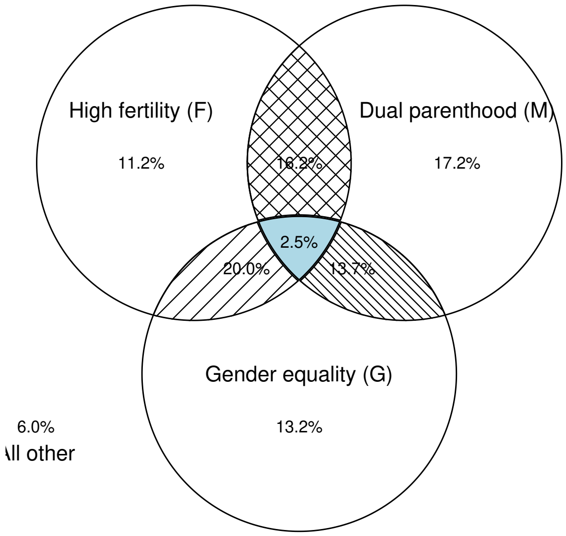
Table 3 reports the mutual information analysis for the three main samples. Across samples, the interaction information is consistently negative, indicating synergy.

	Pairwise MI			Higher-Order		
Sample	$I(F;M)$	$I(F;G)$	$I(M;G)$	Total Corr.	Interaction	Interpretation
OECD	0.043	0.006	0.081	0.165	-0.035^{***}	Synergy
World	0.034	0.099	0.009	0.169	-0.026^{***}	Synergy
U.S. States	0.000	0.082	0.023	0.112	-0.007	Synergy

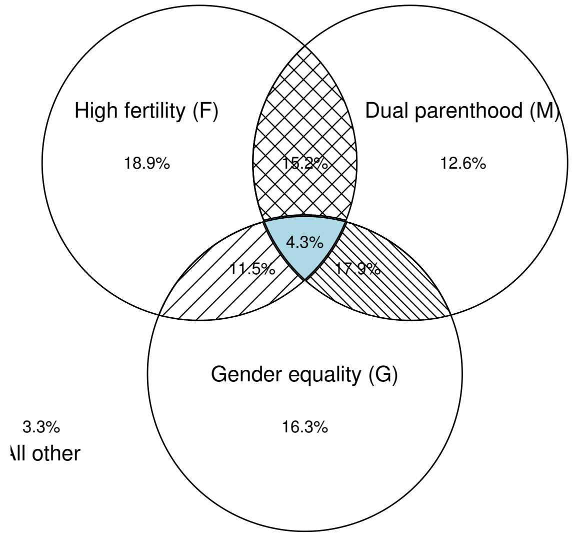
Table 3. Mutual Information Analysis Across Samples

Notes: All values in bits. F = high fertility, M = dual parenthood, G = gender income equality. Pairwise MI = pairwise mutual information. Total Corr. = total correlation (sum of pairwise MI minus joint entropy adjustment). Negative interaction information indicates synergy: conditioning on any variable increases the dependence between the other two. Statistical significance based on bootstrap inference with 5,000 replications: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. See Appendix Table A.1 for detailed results including conditional mutual information, subsample analyses, and U.S. commuting zone results.

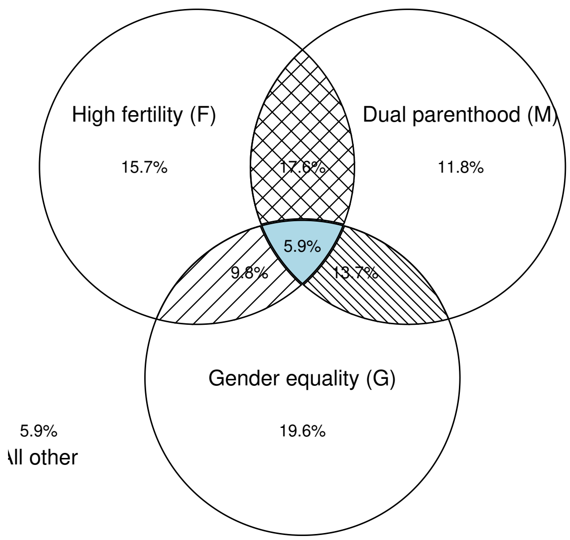
The synergistic pattern is strongest in the cross-country samples. In the OECD sample, the conditional mutual information values substantially exceed the pairwise values: for instance, $I(F;M|G) = 0.078$ bits compared to $I(F;M) = 0.043$ bits unconditionally. This means that the relationship between fertility and dual parenthood becomes



(a) OECD ($N = 721$)



(b) World ($N = 1{,}130$)



(c) U.S. States ($N = 51$)

Figure 1. Three-Way Trade-Off Across Samples

Notes: Venn diagrams showing the joint distribution of high fertility (F), dual parenthood (M), and gender equality (G), each defined at sample-specific medians (see Table 2). Under statistical independence, the central overlap (F+M+G) would contain 12.5% of observations. Results for U.S. commuting zones are reported in Appendix A.5.

stronger once I account for gender equality. The World sample shows a similar pattern, with $I(F;G|M) = 0.125$ bits exceeding $I(F;G) = 0.099$ bits. The U.S. state-level analysis preserves the qualitative finding of synergy, though the magnitude is smaller—likely reflecting the greater homogeneity within a single country. Bootstrap inference confirms that the interaction information is statistically significantly negative in the OECD and World samples

($p < 0.01$), while the U.S. state-level result, though negative, does not reach conventional significance levels due to the smaller sample size.

Transition Patterns

Figure 2 traces OECD countries' transitions across these categories over time. A common pattern emerges: many countries begin with high fertility and dual parenthood (F+M) but progressively move toward gender equality while sacrificing one or both of the other outcomes. The most frequent transition is from F+M to M+G (maintaining dual parenthood while achieving gender equality at the cost of fertility) or to G only (achieving gender equality at the cost of both fertility and dual parenthood).

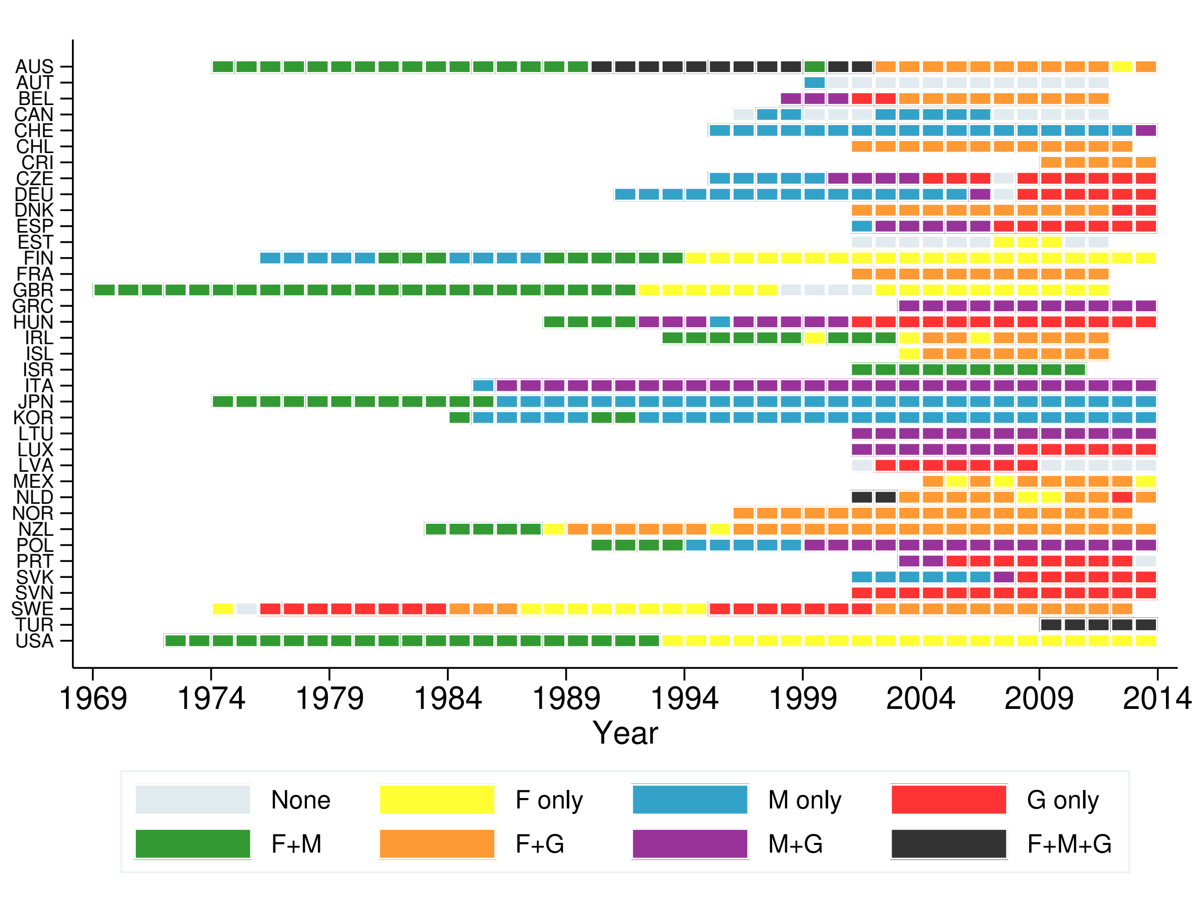


Figure 2. Transition Paths by Country (OECD Sample)

Notes: Temporal transitions in the three categories for each country (F = high fertility, M = dual parenthood, G = gender earnings equality). Countries are labeled by three-letter ISO codes. Data cover 37 OECD countries, 1970–2014. See Appendix Figure A.4 for the corresponding figure for the World sample.

Notably, Australia stands out as one of the few countries that achieved the trinity (F+M+G) for an extended period. As shown in Appendix Table A.2, Australia has among the lowest child-rearing costs relative to per capita GDP among developed nations. This pattern aligns with the theoretical finding in Section 3.4: reducing child-rearing costs is an effective policy lever for alleviating the three-way trade-off.

Geographic Distribution

Figure 3 maps joint attainment across U.S. states. The three states achieving all three outcomes—Alaska, Hawaii, and Minnesota—are geographically and demographically diverse but share relatively high fertility, widespread dual parenthood, and greater gender earnings parity. The geographic dispersion suggests that the trade-off is not driven by regional factors alone.

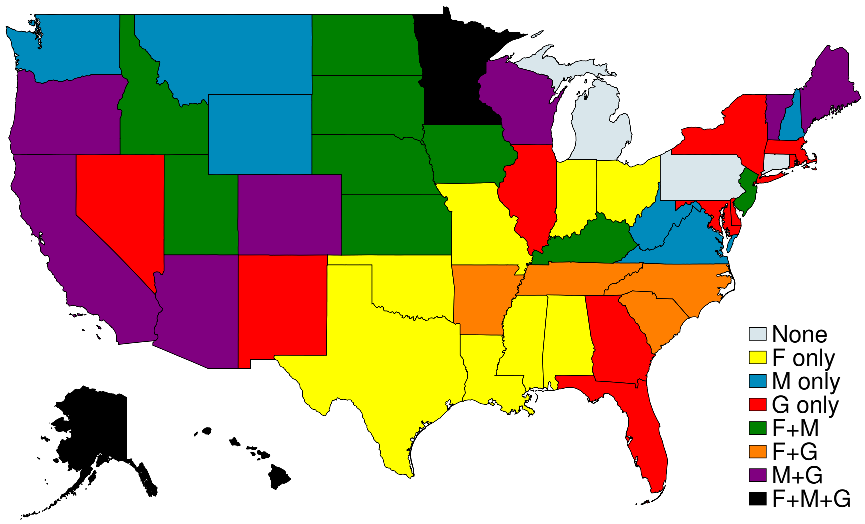


Figure 3. Joint Attainment by U.S. State (2023)

Notes: Map indicating which outcomes—high fertility (F), dual parenthood (M), gender earnings equality (G)—each state achieves, based on median thresholds. See Appendix Figure A.6 for the corresponding map at the commuting zone level.

Robustness

To assess heterogeneity, I stratify samples by income and human capital. Splitting observations at the median GDP per capita and at the median average years of schooling (Barro and Lee, 2013), I redefine indicators using subsample medians in each case. The results, reported in Appendix Figures A.2 and A.3, show that the trade-off persists across both high- and low-income groups and high- and low-human-capital groups. Appendix Table A.1 reports the mutual information analysis for each subsample, confirming that synergy characterizes most subsamples.

2.3 Discussion

The empirical analysis yields two main findings with important implications for understanding the nexus among fertility, family structure, and gender equality.

Finding 1: The Three-Way Trade-Off

Across all samples examined—OECD countries, a global panel of 95 nations, and U.S. states/CZs—simultaneously achieving high fertility, widespread dual parenthood, and gender income equality is rare. The share of observations attaining all three outcomes ranges from 2.5% to 7.6%, consistently below the 12.5% that would be expected under statistical independence. This scarcity is statistically significant and robust to alternative variable definitions, sample restrictions, and levels of geographic aggregation.

Finding 2: Synergistic Dependence Structure

The mutual information analysis reveals that the three outcomes exhibit *synergy* in most samples: the dependence between any two variables increases when conditioning on the third. This finding has a crucial methodological implication. Prior literature has typically examined these outcomes in pairs—fertility and gender gaps, marriage and fertility, or marriage and gender gaps. The synergistic dependence structure documented here indicates that such pairwise analyses miss important information. The tension among fertility, family structure, and gender equality is fundamentally a three-way phenomenon that can only be fully understood by analyzing all three jointly.

Policy Implications

These findings suggest that the three outcomes are not independent policy objectives. Policies targeting one dimension will inevitably affect the others. The theoretical model developed in the next section formalizes these connections and identifies the mechanisms generating the trade-off. Importantly, while this paper documents that the three outcomes are difficult to achieve simultaneously, it does not take a normative stance on whether all three are desirable or how they should be weighted against each other. The empirical patterns documented here delineate a *policy frontier*—the set of feasible combinations—leaving the choice of where to locate on this frontier to policymakers and societies based on their preferences and values.

3 Model

This section develops a unified framework of fertility, family structure, and gender income equality to theorize the empirical patterns documented in the previous section.

3.1 Setup

Individuals are indexed by gender $g \in \{♂, ♀\}$. Each gender comprises half the population. Utility depends on consumption c and fertility n:

$$u(c,n) = \left((1-\beta)\cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}}, \qquad \beta \in (0,1). \tag{1}$$

Following Jones and Schoonbroodt (2010) and Carlos Córdoba and Ripoll (2019), I assume $\rho > 1$ to ensure that consumption and fertility are substitutes and that utility is well-defined for childless individuals, $u(c, 0)$.

To focus on between-gender disparities, wages are uniform within each gender: $w^{♂}$ for men and $w^{♀}$ for women.[1] The gender wage gap at time t is defined as

$$\Gamma^w := \frac{w^{♂}}{w^{♀}}, \tag{2}$$

so a higher Γ^w indicates a larger wage disparity.

Single Individuals

Single men supply one unit of labor inelastically, consume their labor income, and remain childless. Their utility is

$$V^{♂,s} = u(w^{♂}, 0), \tag{3}$$

where the superscript s denotes "single."

Single women, however, can bear children without support from the father.[2] They choose consumption $c^{♀,s}$, labor supply l^s, and fertility n^s to maximize

[1]Incorporating within-gender human capital heterogeneity would require modeling assortative matching (e.g., Low 2014, Chiappori 2017), which is beyond the scope of this paper. Notably, relative to the overall gender earnings gap in the economy, assortative mating does not reduce the within-couple earning differences. Figure A.7 shows that while the gender earnings gap for young workers hover around 10%, the gap among newly-wed spouses without children lie between 20% to 30%.

[2]This is a modeling simplification. In reality, child support enforcement exists, but enforcement is limited: according to the Annie E. Casey Foundation using 2020–2022 Current Population Survey data, only 23% of U.S. female-headed families with children under 18 received any child support in the prior year.

$$V^{\female,s} = \max_{c^{\female,s},l^s,n^s} \quad u(c^{\female,s}, n^s) \tag{4}$$

subject to the budget and time constraints

$$c^{\female,s} = w^{\female} l^s, \quad \text{and} \quad l^s = 1 - \chi n^s,$$

where $\chi > 0$ is the time cost per child. Fertility $n^s \in \mathbb{R}_+$ is continuous, consistent with standard models of female labor supply and fertility (e.g., De La Croix and Doepke 2003). Thus, single women face a simple trade-off between consumption and fertility via endogenous labor supply (Rosenzweig and Wolpin, 1980).

The fertility choice problem in the model is stylized, incorporating two simplifications to enable a transparent analysis of the core mechanism. First, I abstract from childcare marketization (Bar et al., 2018), in which households purchase childcare services in the market as a substitute for parental time. In Section 3.4, however, I examine policies that reduce childcare costs χ for women and demonstrate that such policies can alleviate the three-way trade-off. Second, I do not explicitly model child human capital investment decisions—and thus the quantity-quality trade-off (Becker and Lewis, 1973). Nonetheless, c in the model can be interpreted as encompassing children's human capital. Since consumption c and fertility n are substitutes in both preferences and the budget constraint, this formulation captures the essence of the quantity-quality trade-off.

Married Individuals

In marriage, husbands supply one unit of labor inelastically and transfer a share α of their income to their wives. This transfer captures the economic role of marriage in supporting child-rearing. While individuals take α as given, it is determined in equilibrium (Section 3.2). The share of transfer $\alpha \in [-w_{\male}/w_{\female}, 1]$ so that individuals consume positive amount of consumption. In principle, α could be negative, i.e., husbands could receive transfers from wives.

Husbands consume their remaining income and derive utility from shared fertility n^m and a psychic benefit of marriage λ. Their value is

$$V^{\male,m} = \max_{n^m} \quad u\Big((1-\alpha)w^{\male}, n^m\Big) \cdot \lambda, \tag{5}$$

where the superscript m denotes "married."

Wives choose consumption $c^{\female,m}$, labor supply l^m, and fertility n^m to maximize

$$V^{\female,m} = \max_{c^{\female,m}, l^m, n^m} \quad u(c^{\female,m}, n^m) \cdot \lambda \tag{6}$$

subject to

$$c^{\female,m} = \alpha w^{\male} + w^{\female} l^m, \quad \text{and} \quad l^m = 1 - \chi n^m.$$

Husbands' transfers generate a pure income effect for wives, increasing both consumption and fertility (both normal goods). However, higher female wages induce a substitution effect: since consumption and fertility are substitutes ($\rho > 1$), rising $w^{\female}$ reduces desired fertility—a mechanism emphasized by Greenwood et al. (2005a). Conversely, higher male wages increase fertility via larger transfers. This prediction aligns with Jakobsen et al. (2022), who document these patterns using Danish registry data.

Moreover, since husbands bear no direct child-rearing costs post-transfer, they prefer higher n^m. Wives, bearing the time cost χn^m, prefer lower fertility. Consistent with Doepke and Tertilt (2018), and following Doepke and Kindermann (2019), I assume that fertility in marriage requires mutual consent, so *wives effectively decide marital fertility.*[3]

Marriage Market

The marriage market is competitive. At the start of the period, each individual draws an idiosyncratic taste shock ε for marriage relative to singlehood, following Fréchet distribution with scale θ. By McFadden (1972), the marriage rate for each gender is

$$\mathcal{M}^{\male} = \frac{1}{1 + (V^{\male,s}/V^{\male,m})^{\theta}}, \qquad \mathcal{M}^{\female} = \frac{1}{1 + (V^{\female,s}/V^{\female,m})^{\theta}}. \tag{7}$$

Equilibrium requires

$$\mathcal{M}^{\male} = \mathcal{M}^{\female} = \mathcal{M}, \tag{8}$$

with α and n^m acting as market-clearing "prices."

Unlike Choo and Siow (2006) and related transferable-utility models that use a scalar transfer, this framework abstracts from within-gender heterogeneity but extends the standard setup by allowing a *vector* (α, n^m) to clear the market.

From (7), the equilibrium marriage rate $\mathcal{M}$ is strictly increasing in $V^{\female,m}/V^{\female,s}$—the gains from marriage for each gender.

[3]Assuming men bear no childcare costs is not essential; results hold as long as wives shoulder *more* childcare responsibility—a robust empirical pattern across countries and time (Kleven et al., 2019; Doepke et al., 2023). Section 3.4 extends the model to allow for gender-differentiated childcare costs and derives comparative statics under this more general specification.

Aggregate Variables

Aggregate fertility is the weighted average of marital and non-marital fertility:

$$n = \mathcal{M} \cdot n^m + (1 - \mathcal{M}) \cdot n^s. \tag{9}$$

The share of children living with both parents is

$$\mathcal{D} = \frac{\mathcal{M} \cdot n^m}{n}. \tag{10}$$

Average female labor supply is

$$l^{\female} = \mathcal{M} \cdot l^m + (1 - \mathcal{M}) \cdot l^s = 1 - \chi n. \tag{11}$$

Labor income is $y^{\male} = w^{\male}$ for men and $y^{\female} = w^{\female} l^{\female}$ for women, so the gender income gap is

$$\Gamma^y = \frac{y^{\male}}{y^{\female}} = \frac{\Gamma^w}{l^{\female}}. \tag{12}$$

Equation (12) shows that the gender income gap widens if the wage gap Γ^w increases or if women reduce labor supply (i.e., $l^{\female}$ falls due to higher fertility).

3.2 Equilibrium Characterization

The equilibrium of the economy is characterized in three steps.

First, for a given marriage rate $\mathcal{M}$, men's optimal choice yields α as a function of marital fertility n^m:

$$\alpha = 1 - \left[\left(\left(\frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{\frac{1}{\theta}} \cdot \frac{1}{\lambda} \right)^{\frac{\rho-1}{\rho}} - \frac{\beta}{1 - \beta} \cdot \left(\frac{n^m}{w^{\male}} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}. \tag{13}$$

Since $\rho > 1$, the implicit function theorem implies that $\alpha(n^m)$ is strictly increasing and concave in n^m. It equals zero when $n^m = 0$. Moreover, for fixed n^m, $\alpha(n^m)$ decreases with $w^{\male}$: higher male income allows a smaller income share to sustain any given level of marital fertility.

On the other side of the marriage market, the first-order condition for married women determines n^m as a function of α:

$$n^m \cdot \left[\left(\frac{(1 - \beta)\chi}{\beta} \right)^{\rho} \cdot (w^{\female})^{\rho - 1} + \chi \right] = 1 + \alpha \Gamma^w. \tag{14}$$

Equation (14) shows that $n^m(\alpha)$ is linear and increasing in the transfer α. Even when $\alpha = 0$, $n^m > 0$ (matching single women's positive fertility). The function shifts downward with $w^{♀}$, confirming that the substitution effect dominates the income effect as female wages rise.

Lastly, I make an additional assumption, which serves as a sufficient condition to ensure the uniqueness of the marriage market equilibrium.

Assumption 1. *The psychic benefit of marriage λ satisfies*

$$\lambda \geq \left(\frac{1-\beta}{\beta}\right)^{\frac{1}{\rho-1}} \cdot \frac{1}{w^{♂}\chi} \cdot \left[\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho} (w^{♀})^{\rho-1} + \chi\right]. \tag{15}$$

This condition ensures that the marginal man in equilibrium does not require compensation (i.e., $\alpha \geq 0$) even if his wife remains childless.

The bound in Assumption 1 depends only on primitives $(\beta, \rho, \chi, w^{♂}, w^{♀})$ and ensures that marriage provides sufficient non-pecuniary benefits to clear the market with non-negative transfers. In the calibration, the estimated $\lambda = 1.03$ satisfies this condition for all periods.

Under this assumption, the marriage market equilibrium is fully characterized by the following results.

Lemma 1. *Under Assumption 1, for any wage pair $\{w^{♂}, w^{♀}\}$, there exists a unique fixed point $(\alpha, n^m) \in (0,1) \times \mathbb{R}_{++}$ that clears the marriage market.*

Proof: See Appendix C.2.

Lemma 2. *The gains from marriage for women are $V^{♀,m}/V^{♀,s} = \lambda \cdot (1 + \alpha\Gamma^w)$.*

Proof: See Appendix C.3.

Lemma 2 shows that the equilibrium marriage rate $\mathcal{M}$ depends on the psychic benefit λ and the *economic* gains from marriage for women, $\alpha\Gamma^w$—the product of the transfer share and the gender wage gap. This result–that in a setting without heterogeneities within gender, marriage rate is increasing in gender wage gap–echoes the classic result in Becker (1973).

Combined with Equation (34) in the Appendix, Lemma 2 implies that the share of dual parenthood $\mathcal{D}$ (defined in (10)) is strictly increasing in $\mathcal{M}$. Hence, $\mathcal{M}$ and $\mathcal{D}$ are interchangeable in the analysis that follows.

3.3 Three-Way Trade-Off

The equilibrium relationships among aggregate fertility n, marriage rate $\mathcal{M}$, female labor supply $l^{♀}$, and gender income gap Γ^y are summarized by three equations:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta}, \tag{16}$$

$$\Gamma^y = \frac{\Gamma^w}{l^{♀}}, \tag{17}$$

$$l^{♀} = 1 - \chi n. \tag{18}$$

These equations reveal the model's core tensions. Equation (16) shows that higher gender wage gaps Γ^w support higher marriage rates. But (17) implies that a large Γ^w undermines gender income equality unless female labor supply is high. Yet (18) shows that high female labor supply $l^{♀}$ requires low fertility.

Thus, for fixed parameters, the model establishes a *three-way trade-off* among high fertility, high dual-parenthood (or marriage), and gender income equality, mirroring the empirical patterns documented in Section 2. This is shown by considering the three pairwise combinations:

(1) *High fertility and high dual-parenthood*: High n implies low $l^{♀}$ via (18). To sustain high $\mathcal{M}$, Γ^w cannot be too low via (16). Thus, Γ^y is necessarily high via (17).

(2) *High fertility and gender income equality*: High n again implies low $l^{♀}$. To achieve low Γ^y, Γ^w must be very small via (17). Because α is bounded above by one, a very small wage gap reduces $\mathcal{M}$ via (16).[4]

(3) *High dual-parenthood and gender income equality*: Because α is bounded above, high $\mathcal{M}$ requires large Γ^w via (16). To offset this and achieve low Γ^y, $l^{♀}$ must be high via (17). Thus, fertility n is low via (18).

3.4 Policy Remedies

The trade-off documented in this paper has profound policy implications. Many governments pursue policies to boost fertility, strengthen family structure, or narrow gender income gaps—often treating these as independent objectives. This analysis shows, however, that the three outcomes are deeply interconnected. Policies targeting one dimension inevitably affect the others. I therefore use the model to evaluate the equilibrium effects of common policy approaches and derive rigorous comparative statics results.

[4] In reality, the upper bound of transfer α could be much smaller than one due to social norms or subsistence constraints.

Pro-Marriage Policies

Pro-marriage policies—such as joint income taxation—are often motivated not only by cultural or religious values but also by evidence that family structure disparities contribute to persistent gaps in child human capital (Kearney 2023) and income inequality across demographic groups, e.g., black versus white (Autor et al., 2019). In the model, such policies increase the psychic benefit of marriage λ.

Proposition 1. *An increase in the psychic benefit of marriage λ raises aggregate fertility n and the marriage rate $\mathcal{M}$, but widens the gender income gap Γ^y.*

Proof: See Appendix C.4.

The intuition is straightforward. A higher λ raises the marriage rate $\mathcal{M}$ without altering marital fertility n^m. Since $n^m > n^s$, aggregate fertility n increases via a composition effect. This reduces female labor supply and widens the gender income gap Γ^y.

Gender Equality Policies

Gender equality policies—such as anti-discrimination laws and pay equity mandates—aim to reduce labor market misallocation and boost aggregate productivity (Hsieh et al. 2019), alongside promoting female empowerment. In the model, these correspond to an increase in $w^{♀}$ with $w^{♂}$ fixed.

Proposition 2. *An increase in the female wage $w^{♀}$ (holding $w^{♂}$ fixed) reduces aggregate fertility n, lowers the marriage rate $\mathcal{M}$, and narrows the gender income gap Γ^y.*

Proof: See Appendix C.5.

Higher female wages reduce both single and marital fertility (n^s and n^m) due to substitution between consumption and children. Lower n^m decreases the equilibrium transfer α, which reduces the marriage rate $\mathcal{M}$. The resulting shift toward singlehood further lowers aggregate fertility n. However, female labor supply rises, and combined with higher wages, the gender income gap Γ^y narrows.

Pro-Fertility Policies: Reducing Child-Rearing Costs

Pro-fertility policies—such as baby bonuses, public childcare, or initiatives promoting more equal division of childcare responsibilities between genders—are increasingly adopted to counter population aging, which threatens pension sustainability (Bongaarts 2004) and long-run growth (Jones 2022). In the baseline model, these reduce the time cost of child-rearing χ.

Proposition 3. *A reduction in the child-rearing cost χ increases aggregate fertility n, raises the marriage rate $\mathcal{M}$, and widens the gender income gap Γ^y.*

Proof: See Appendix C.6.

A lower χ increases both n^s and n^m. Higher marital fertility raises the equilibrium transfer α and marriage rate $\mathcal{M}$, generating a reinforcing composition effect that further boosts aggregate fertility n. However, female labor supply *decreases* because total childcare time χn rises: the indirect effect of higher fertility dominates the direct effect of lower χ. Thus, the gender income gap Γ^y widens.

Summary of Baseline Policy Effects

Table 4 summarizes the comparative statics results for the three policy instruments.

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{♀}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

Table 4. Summary of Policy Effects

Notes: Signs indicate the direction of change in each outcome variable in response to the indicated policy shift.

The results reveal a stark conclusion: none of the three policy instruments can mitigate the three-way trade-off. Pro-marriage policies and pro-fertility policies both increase fertility and marriage but widen gender gaps. Gender equality policies narrow the gap but reduce fertility and marriage. The trade-off appears fundamental within the baseline model.

One additional caveat applies to pro-fertility policies. Their fiscal cost is proportional to female wages. As economies grow and gender discrimination falls—raising $w^{♀}$—these policies become more expensive. To sustain a given fertility level, their cost rises as a share of GDP. The intuition is straightforward: the government subsidizes a sector (child-rearing) whose output is substitutable to market goods and has seen no productivity growth.

Redistributing Childcare Costs Between Spouses

The baseline model assumes that only women bear the time cost of child-rearing. While this assumption is grounded in empirical evidence—time-use surveys consistently show that mothers spend substantially more time on childcare than fathers across virtually all countries (Aguiar and Hurst, 2007; Kleven et al., 2019)—the division of childcare labor is not immutable. It responds to economic incentives, social norms, and policy interventions such as paternity leave mandates and flexible work arrangements (Farré and González, 2024).

To explore the implications of altering the gender distribution of childcare responsibilities, I extend the model to allow for gender-differentiated childcare costs within marriage. Let $\chi^{♀}$ denote the time cost per child borne by wives and $\chi^{♂}$ the cost borne by husbands, where $\chi^{♀} > \chi^{♂}$ (women shoulder a larger share). Single mothers continue to bear the full cost χ when raising children alone, since they have no partner with whom to share responsibilities.

In this extended setup, the wife's budget constraint within marriage becomes

$$c^{♀,m} = \alpha w^{♂}(1 - \chi^{♂} n^m) + w^{♀}(1 - \chi^{♀} n^m), \tag{19}$$

while the husband's budget constraint becomes

$$c^{♂,m} = (1 - \alpha) w^{♂}(1 - \chi^{♂} n^m). \tag{20}$$

Total household income is $w^{♂}(1-\chi^{♂} n^m)+w^{♀}(1-\chi^{♀} n^m)$. Redistributing childcare from wives to husbands—increasing $\chi^{♂}$ and decreasing $\chi^{♀}$ by equal amounts—has two opposing effects on household income. Holding fertility fixed, it reduces total income because the husband's time is more valuable ($w^{♂} > w^{♀}$). However, it also affects equilibrium fertility, which feeds back into income through labor supply. The net effect depends on parameter values.

The full equilibrium conditions and derivations appear in Appendix D. Here I characterize the conditions under which redistribution raises fertility and marriage.

Proposition 4. *Consider the extended model with gender-differentiated childcare costs. Suppose that (i) $\chi^{♀} > \chi^{♂}$, (ii) $w^{♂} > w^{♀}$, and (iii) $\rho > 1$. Define the wife's effective childcare cost as*

$$\tilde{\chi}^{♀} \equiv \frac{\alpha w^{♂} \chi^{♂} + w^{♀} \chi^{♀}}{\alpha w^{♂} + w^{♀}},$$

which is the income-weighted average of the two spouses' childcare costs from the wife's perspective. Consider a redistribution that increases $\chi^{♂}$ and decreases $\chi^{♀}$ by equal amounts $\delta > 0$, holding $\chi^{♂} + \chi^{♀}$ and χ fixed.

(i) The wife's effective childcare cost $\tilde{\chi}^{♀}$ decreases if and only if

$$\alpha < \frac{w^{♀}}{w^{♂}}. \tag{21}$$

(ii) When condition (21) holds, marital fertility n^m increases. Single fertility n^s is unchanged.

(iii) When condition (21) holds, the wife's utility from marriage rises relative to singlehood.

(iv) The husband's utility from marriage rises if and only if

$$\frac{\beta}{1-\beta} \cdot \left(\frac{c^{\male,m}}{n^m}\right)^{\frac{1}{\rho}} > (1-\alpha)w^{\male} \cdot \left(\frac{n^m}{\partial n^m/\partial \delta} + \chi^{\male}\right), \tag{22}$$

where $\partial n^m/\partial \delta > 0$ is the fertility response to redistribution. This condition is more likely to hold when (a) the preference weight on children β is large, (b) the husband's initial childcare share $\chi^{\male}$ is small, (c) the elasticity of substitution ρ is low, and (d) the fertility response is strong.

(v) When conditions (21) and (22) both hold, both spouses prefer marriage at higher rates, so the marriage rate $\mathcal{M}$ increases.

(vi) Aggregate fertility n increases whenever marital fertility rises and the marriage rate does not decline.

Proof: See Appendix D.4.

The central intuition operates through three channels. First, condition (21) ensures that redistributing childcare to husbands lowers the wife's effective cost of children. This occurs when the transfer she receives from her husband ($\alpha w^{\male}$) is smaller than her own earnings ($w^{\female}$), so that shifting the burden toward the income source with smaller weight in her budget reduces her effective cost. When this condition holds, marital fertility rises.

Second, when condition (21) holds, the wife's overall gains from marriage unambiguously increase. The proof in Appendix D.4 demonstrates that the product of marital fertility and the wife's utility multiplier is strictly decreasing in her effective childcare cost. Thus, any reduction in $\tilde{\chi}^{\female}$ raises the wife's utility from marriage relative to singlehood. Importantly, this result holds for all $\rho > 1$—no additional restriction on the elasticity of substitution is required.

Third, condition (22) characterizes when the husband also benefits. Redistribution creates two opposing effects for the husband: higher marital fertility n^m increases his utility from children, but his consumption $c^{\male,m}$ falls because he spends more time on childcare. The left-hand side of (22) captures the marginal utility gain from children (scaled by the consumption-fertility ratio), while the right-hand side captures the marginal utility cost from reduced consumption. The husband benefits when the fertility gain dominates.

Condition (22) is more likely to hold in economies where: (a) preferences place substantial weight on children (β large); (b) fathers initially contribute little to childcare ($\chi^{\male}$

small), so the direct consumption loss is limited; (c) consumption and children are not highly substitutable (ρ close to 1), so the marginal utility of additional children is high; and (d) the fertility response to redistribution is strong, which occurs when condition (21) holds with slack. When both conditions are satisfied, the redistribution is Pareto-improving within the household.

These results suggest that policies targeting the intra-household division of labor—such as paternity leave mandates or "use-it-or-lose-it" parental leave provisions—can potentially raise fertility and marriage while promoting gender equality. The key requirements are that (i) wives' own labor earnings exceed the transfers they receive from husbands, and (ii) the husband's utility gain from higher fertility outweighs his consumption loss. Countries like Sweden and Iceland, which have implemented strong paternity leave policies with reserved quotas for fathers, provide natural testing grounds for these predictions.

One caveat applies: I have abstracted from potential adjustment costs or efficiency losses if fathers are initially less productive caregivers than mothers. Incorporating such frictions would temper the quantitative magnitudes.

4 Quantitative Assessment

To illustrate the quantitative implications of the unified framework, this section calibrates the model using Mexico's economic and demographic transition from 1990 to 2015. Mexico is chosen for three main reasons. First, it provides a long and consistent time series of high-quality data. Second, during this period, Mexico experienced rapid GDP growth accompanied by a complete reversal of the gender education gap, making it an ideal case for studying the effects of gender-neutral and gender-biased technological changes on fertility, family structure, and income inequality. Third, many other countries have followed similar trajectories (see Appendix Figure A.1). Thus, the insights gained from the Mexican case are highly relevant to a broader set of economies.

4.1 Data and Calibration

I draw the Mexican data from the world sample. Figure 4 plots Mexico's transition path. Fertility declines sharply from 3.5 to approximately 2.1 children per woman. The share of children living with both parents falls from 81% to 75%, while the share with only the mother present rises from 11% to 19%. Gender income equality improves as women's labor income share increases from 24% to 32%. Over the same period, GDP per capita rises from about \$12,000 to over \$18,000 (in 2017 US dollars). Most strikingly, the gender education gap reverses: in 1990, women aged 25–35 had 0.8 fewer years of schooling than men; by 2010, the gap closed; and by 2015, women in this age range had 0.1 more years of schooling.

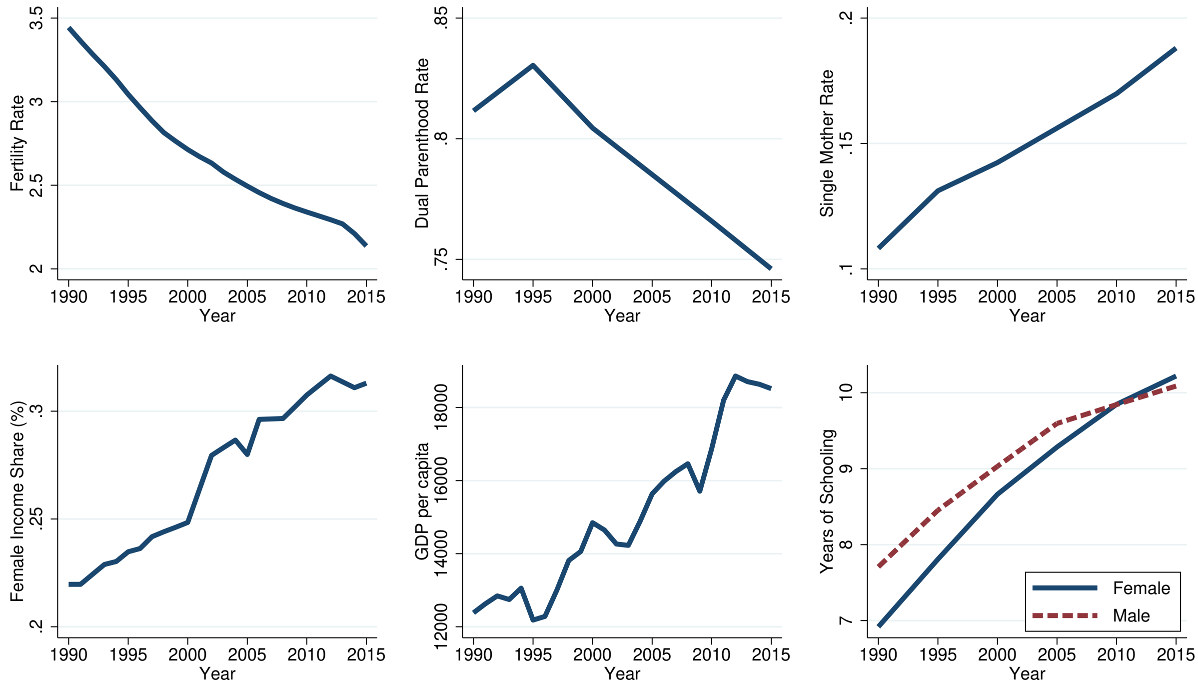


Figure 4. Mexico's Transition Path (1990–2015)

Notes: Fertility is total children per woman (United Nations). Dual parenthood is the share of children living with both biological parents; single-mother share is the share living only with mother (Brenøe and Wasserman 2025). Female labor income share is from the World Inequality Database. GDP per capita is in 2017 US dollars (OECD). Gender education gap is the difference in average years of schooling for ages 25–35 (Barro and Lee 2013).

The model is calibrated by solving for equilibrium year by year. I assume that time-varying wages $\{w_t^{♀}, w_t^{♂}\}$ are the sole exogenous drivers. I further decompose these wage trends into three components:

- $A_t = w_t^{♂}$: gender-neutral productivity,
- Γ_t^h: gender gap in human capital,
- $B_t = \frac{w_t^{♂}/w_t^{♀}}{\Gamma_t^h}$: gender-biased productivity.

Calibration proceeds in three steps. First, the time cost of children is fixed at $\chi = 0.075$ following De La Croix and Doepke (2003). The human capital gender gap Γ_t^h is constructed from average years of schooling using the development accounting framework of Caselli (2005) which utilizes Mincerian return estimates. Second, female labor supply $l_t^{♀}$ is inferred from observed fertility via Equation (18). Wages $\{w_t^{♀}, w_t^{♂}\}$ are then backed out using GDP per capita and the female labor income share from Figure 4. Finally, the preference and marriage parameters $\{\beta, \rho, \lambda, \theta\}$ are jointly calibrated to match the observed time paths of fertility and dual parenthood.

Identification of these parameters is intuitive:

(1) β governs the weight on fertility and targets average fertility levels;

(2) ρ determines substitutability between consumption and children and targets the slope of fertility decline as productivity rises;

(3) λ captures the psychic benefit of marriage and targets the average dual-parenthood share;

(4) θ governs dispersion in marriage taste shocks and targets the responsiveness of dual parenthood to changes in fertility and gender wage gaps.

Table 5 reports the calibrated values. The elasticity of substitution $\rho = 1.50$ implies that rising wages reduce fertility as substitution effects dominates income effects. The psychic benefit of marriage $\lambda = 1.03$ implies a 3% utility premium to marriage beyond resource exchange via (α, n^m).

	Interpretation	Value	Target	Source
χ	time cost of children	0.075	De La Croix and Doepke (2003)	
A_t	gender-neutral productivity		GDP per capita	OECD
B_t	gender-biased productivity		female labor income share	World Inequality Database
Γ_t^h	gender human capital gap		years of schooling	Barro and Lee (2013)
β	preference weight on n	0.104	fertility level	United Nations
ρ	c–n substitutability	1.50	fertility trend	United Nations
λ	psychic benefit of marriage	1.03	dual parenthood level	Brenøe and Wasserman (2025)
θ	taste shock dispersion	3.40	dual parenthood trend	Brenøe and Wasserman (2025)

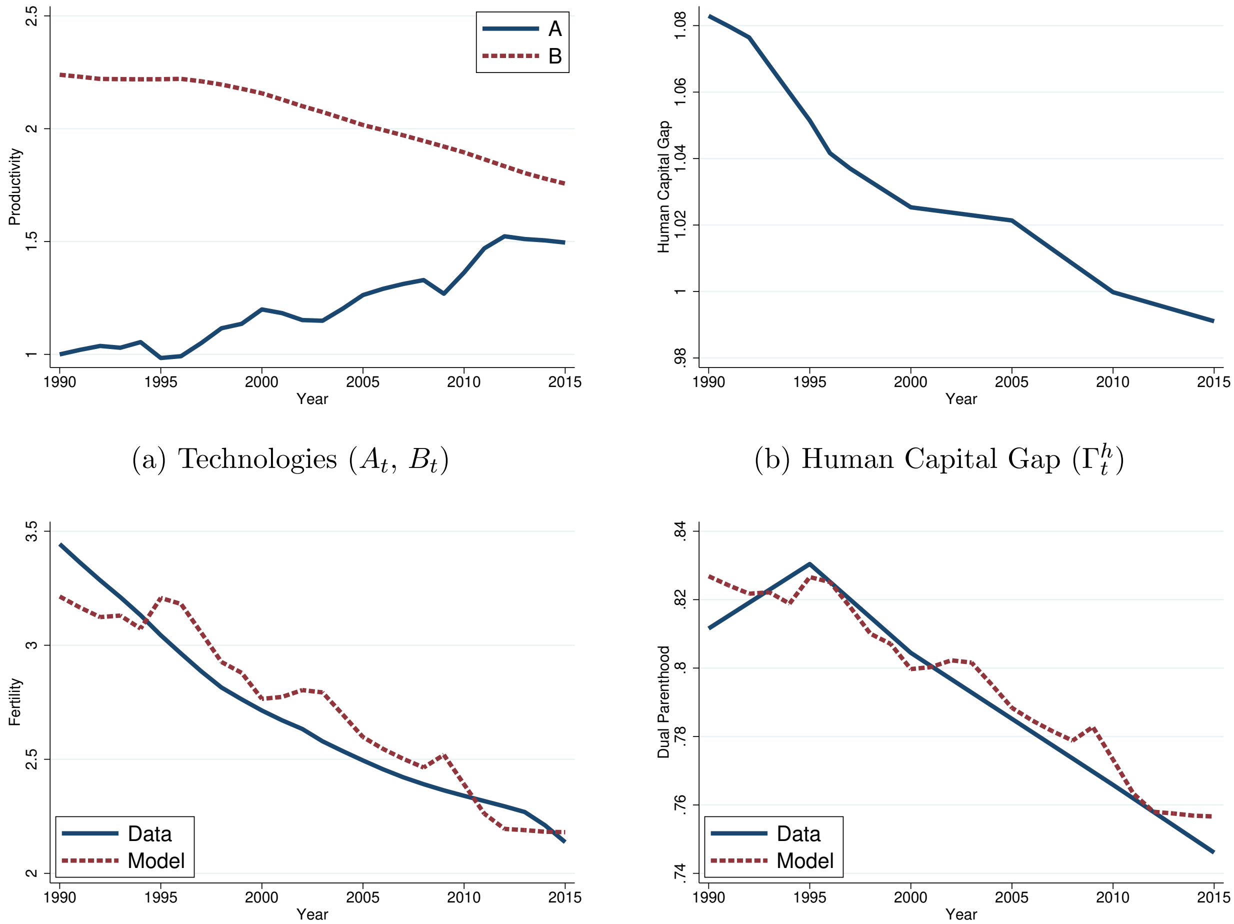
Table 5. Calibration Results

Notes: Time-varying inputs (A_t, B_t, Γ_t^h) are constructed directly from data. Preference and marriage parameters minimize the distance between model and data for fertility and dual parenthood over 1990–2015.

Figure 5 displays the calibrated driving forces (A_t, B_t, Γ_t^h) and compares model predictions with data. Gender human capital convergence ($\Gamma_t^h \to 1$) begins in 1990. The decline in gender-biased productivity B_t starts in the late 1990s. Gender-neutral productivity A_t rises steadily after the 1994 Peso Crisis. With only four calibrated parameters, the model closely tracks both fertility and dual-parenthood trends.

4.2 Equilibrium Variables

One advantage of the framework here is that it yields estimates of within-marriage transfers and gender-specific utility levels across family structures—quantities that are difficult, if not



(a) Technologies (A_t, B_t)

(b) Human Capital Gap (Γ_t^h)

(c) Fertility

(d) Dual Parenthood

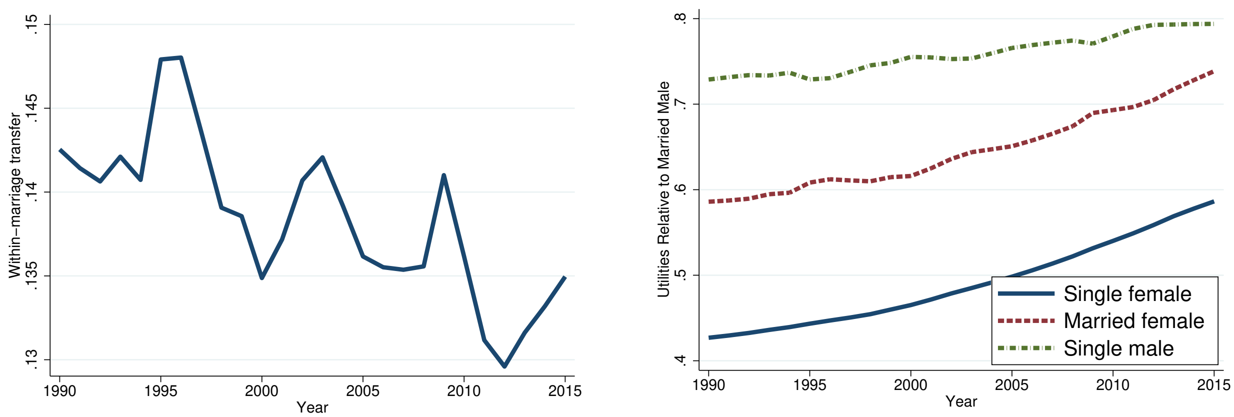
Figure 5. Model Fit: Mexico (1990–2015)

Notes: Solid lines: data. Dashed lines: model. Fertility is total children per woman (United Nations). Dual parenthood is the share of children living with both biological parents (Brenøe and Wasserman 2025). $A_t = w_t^{♂}$ is gender-neutral productivity (from GDP per capita, OECD). $B_t = (w_t^{♂}/w_t^{♀})/\Gamma_t^h$ is gender-biased productivity (from female labor income share, World Inequality Database). Γ_t^h is the gender human capital gap (from years of schooling, Barro and Lee 2013). Model is calibrated using parameters in Table 5.

impossible, to observe directly in data yet crucial for linking income inequality to welfare inequality.

Figure 6 reports key equilibrium outcomes from the calibrated model. Figure 6a shows that within-marriage income transfers α_t fall by 1 percentage point as marital fertility declines. This reduction is small relative to the sharp drop in n^m, because the narrowing gender wage gap requires husbands to transfer a *larger share* of their *smaller relative income* to sustain fertility via income effects.

Figure 6b reveals that relative to married men, the utilities of single men rise. Despite lower within-marriage transfers, married men are relatively worse off due to lower within-marriage fertility. On the other hand, both single and married women benefit from wage convergence. For example, while single women's utilities were initially 43% of married men's in 1990, it rises to 59% by 2015. Notably, even though the transfers from husbands to wives decline, the husband–wife welfare gap narrows substantially from 42% to 26%. The decomposition results in the next section reveals that this result is driven by women's rising earning power.



(a) Within-Marriage Transfers (α_t)

(b) Relative Utilities (Benchmark: $V^{♂,s}$)

Figure 6. Model-Generated Equilibrium Outcomes

Notes: Model simulations for Mexico, 1990–2015, using calibrated parameters from Table 5. Panel (a) plots the share of male income transferred to wives within marriage (α_t). Panel (b) reports utilities of married men ($V^{♂,m}$), married women ($V^{♀,m}$), and single women ($V^{♀,s}$) relative to single men ($V^{♂,s}$).

Importantly for the policy analysis, the equilibrium transfer $\alpha \approx 0.10$ satisfies condition (21) throughout the sample period. Since $w^{♀}/w^{♂}$ ranges from approximately 0.45 in 1990 to 0.62 in 2015, the condition $\alpha < w^{♀}/w^{♂}$ holds comfortably. This suggests that policies redistributing childcare from wives to husbands would benefit wives in Mexico's economic environment.

To assess whether the husband would also benefit from redistribution, I evaluate condition (22) at the calibrated equilibrium values. Throughout the sample period, the left-hand side of (22) exceeds the right-hand side, indicating that the husband's utility gain from higher fertility would outweigh his consumption loss from increased childcare responsibilities. This reflects Mexico's relatively high preference weight on children ($\beta = 0.104$) and the assumption that fathers initially bear little childcare burden ($\chi^{♂} \approx 0$). Together with the satisfaction of condition (21), these results suggest that policies redistributing childcare from mothers to fathers could be Pareto-improving in Mexico's economic environment.

4.3 Decomposition

Another advantage of the model is that it can be used to conduct decompositions to better understand the fundamental drivers of the equilibrium outcomes. Here, I decompose changes in fertility n_t, dual parenthood $\mathcal{D}_t$, gender income gap Γ_t^y, and spousal welfare gap $V_t^{♀,m}/V_t^{♂,m}$ into contributions from rising gender-neutral productivity A_t, falling gender-biased productivity B_t, and converging human capital gap Γ_t^h.[5]

I conduct the decomposition in a sequential order:

(1) Vary only A_t, holding B_t and Γ_t^h at 1990 levels.

(2) Allow A_t and B_t to vary, keeping Γ_t^h fixed.

(3) Allow all three factors to evolve.

Figure 7 presents the results. Rising A_t accounts for approximately 50% of the fertility decline: higher opportunity costs reduce childbearing. It also explains approximately 50% of the drop in dual parenthood, as falling marital fertility n^m reduces men's gains from marriage, shifting births to single mothers. It contributes about 10% to gender income gap convergence via higher female labor supply. It also contributes negatively to the spousal welfare gap because within-marriage transfers decline substantially as fertility falls.

Declining B_t explains approximately 40% of fertility decline, approximately 40% of dual-parenthood decline, and the majority of gender income and spousal welfare convergence. This operates through two channels: (i) higher absolute female wages reduce fertility and marriage while increasing female labor supply and their relative consumption; (ii) rising relative female wages require larger transfers α to sustain n^m which lowers equilibrium marriage rates.

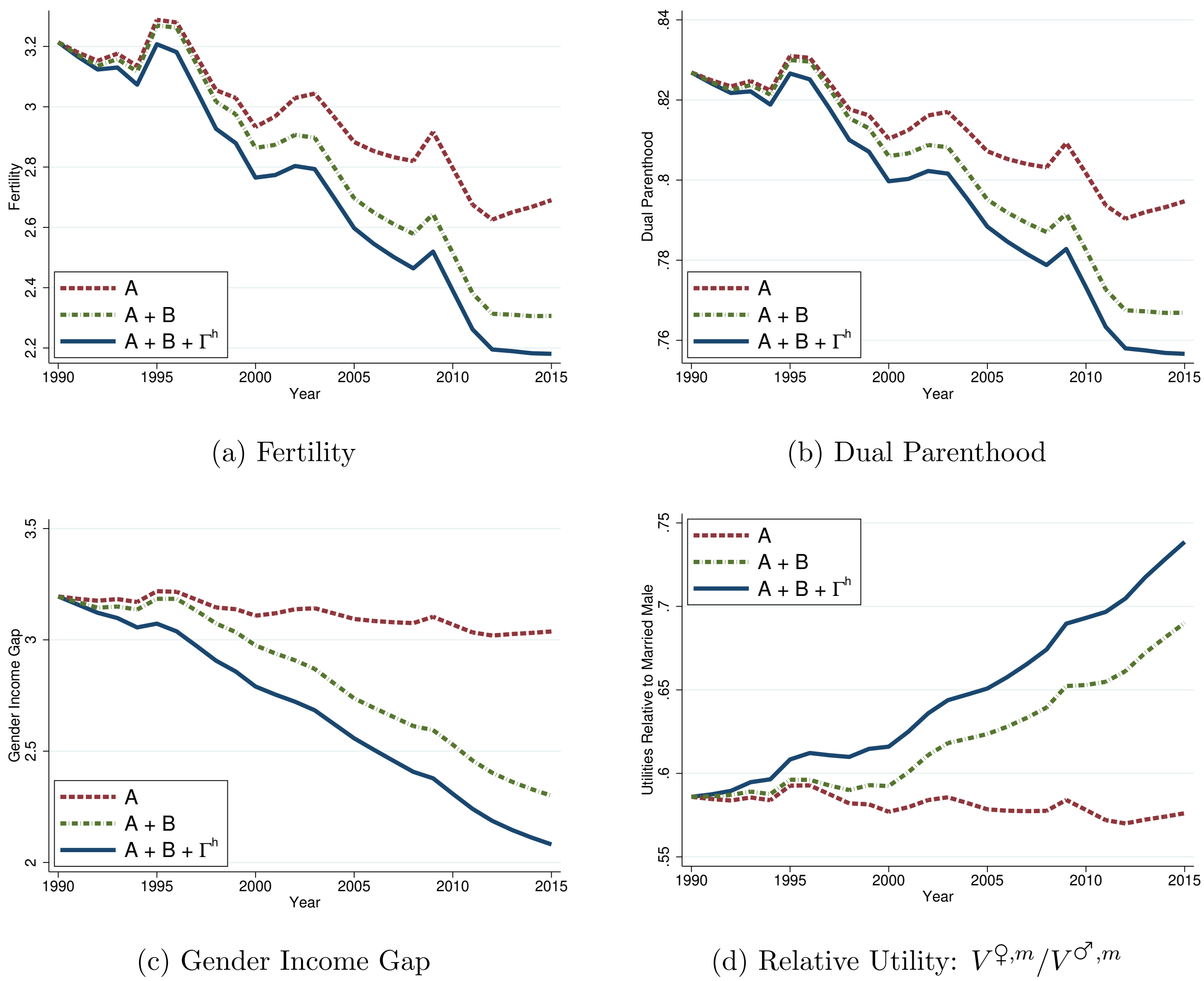
The reversal in Γ_t^h accounts for the remainder. By construction, B_t absorbs residual gender wage gaps after Γ_t^h is accounted for, so their relative contributions reflect changes in gender-specific schooling in the data.

4.4 Prediction

Lastly, I use the calibrated model to forecast fertility, dual parenthood, and the gender income gap to 2040. Gender-specific wages $\{w_t^{♀}, w_t^{♂}\}$ are linearly extrapolated from 1990–2015 trends. The model is then solved period-by-period using fixed parameters.

Figure 8 shows the results. Continued wage growth—especially for women—drives fertility to approximately 1.5 children per woman by 2040. Dual parenthood falls from 75% in 2015 to approximately 70%. The gender income gap continues to narrow, reaching approximately 60% of male income (down from 220% in 1990). Fertility data through 2023 provide

[5]Figure A.8 presents additional results on the decomposition of $V_t^{♀,s}/V_t^{♂,m}$ and $V_t^{♂,s}/V_t^{♂,m}$.



(a) Fertility

(b) Dual Parenthood

(c) Gender Income Gap

(d) Relative Utility: $V^{♀,m}/V^{♂,m}$

Figure 7. Decomposition of Mexico's Transition (1990–2015)

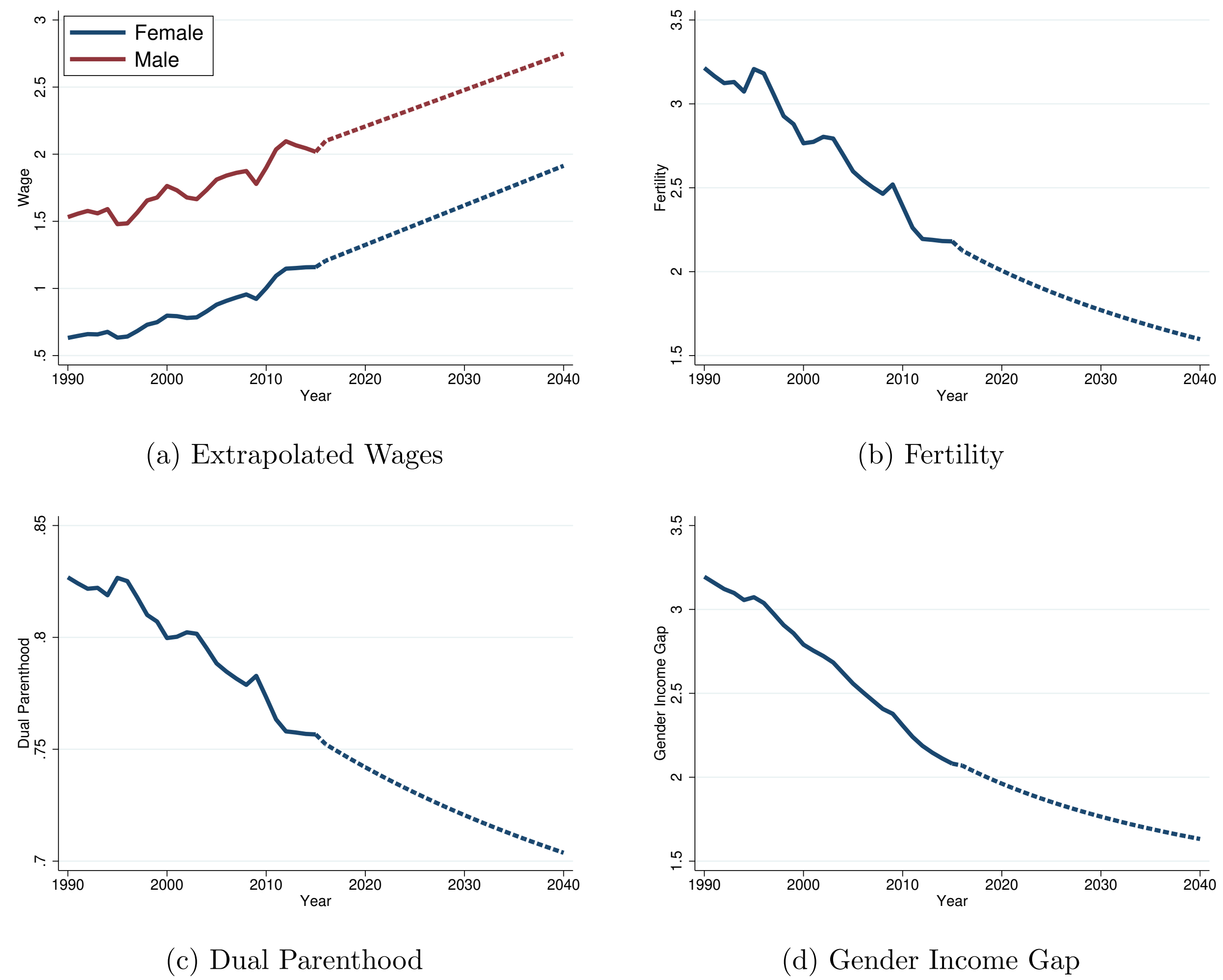
Notes: Solid line: data. Dashed lines: counterfactual paths. "A_t only": vary gender-neutral productivity, fix B_t, Γ_t^h at 1990. "$A_t + B_t$": allow both to vary, fix Γ_t^h. "All": full model.

an out-of-sample check: the model predicts 1.93 children per woman; actual fertility rate is 1.91.

This exercise underscores the model's value. Rather than extrapolating fertility, marriage, or gender gaps in isolation, it treats them as joint endogenous outcomes of gender-specific wage dynamics. Decomposing wages into A_t, B_t, and Γ_t^h enables refined forecasts under alternative assumptions about technological or educational convergence.

5 Dynamic Extension

This section introduces a dynamic extension in which the gender human capital gap evolves endogenously with the marriage rate. Specifically, the law of motion is



(a) Extrapolated Wages (b) Fertility

(c) Dual Parenthood (d) Gender Income Gap

Figure 8. Model-Based Predictions: Mexico to 2040

Notes: Dashed lines: linear extrapolation of gender-specific wages from 1990–2015. Solid lines: model predictions using calibrated parameters (Table 5). Fertility in 2023: model predicts 1.93, actual is 1.91 (United Nations).

$$\Gamma^h_{t+1} = \mathcal{H}(\mathcal{M}_t) \qquad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0. \tag{23}$$

I adopt a linear specification:

$$\mathcal{H}(\mathcal{M}_t) = \gamma_0 + \gamma_1 \mathcal{M}_t, \qquad \gamma_1 > 0, \tag{24}$$

where γ_0 captures the baseline human capital gap when all children are raised by single mothers, and γ_1 measures the additional human capital advantage boys gain from dual-parent households relative to girls. This linear form provides a tractable benchmark while capturing the key mechanism.

The key assumption $\mathcal{H}'(\mathcal{M}_t) = \gamma_1 > 0$ is motivated by a large empirical literature documenting that growing up without both biological parents harms boys more than girls (e.g., Sommers 2001, Bertrand and Pan 2013, Chetty et al. 2016, Autor et al. 2019, Wasserman 2020, Reeves 2022, Frimmel et al. 2024). This gender-differentiated impact may arise from (1) same-gender role model effects, (2) differential sensitivity to parental inputs, or (3) varying exposure to alternative institutions such as schools or neighborhoods.

While the existing studies mostly focus on the U.S. setting, the measured magnitude of this channel is economically significant. For instance, Autor et al. (2019) show that racial differences in single motherhood explain much of the black–white gender gap in outcomes. Autor et al. (2023) find that family structure, via socioeconomic status, accounts for a substantial share of gender disparities in high school performance.

With the linear specification (24), the dynamic system has well-defined stability properties. To characterize these, I first derive the response of marriage to the human capital gap.

Lemma 3. *In equilibrium, the derivative of the marriage rate with respect to the gender human capital gap is*

$$\frac{\partial \mathcal{M}}{\partial \Gamma^h} = \frac{\theta \mathcal{M}(1-\mathcal{M}) \cdot \alpha}{1+\alpha \Gamma^w} \cdot \frac{\partial \Gamma^w}{\partial \Gamma^h} > 0, \tag{25}$$

where $\partial \Gamma^w / \partial \Gamma^h = B > 0$ (the gender-biased productivity factor).

Proof: See Appendix E.

Combining the law of motion (24) with Lemma 3, local stability around a steady state $(\mathcal{M}^*, \Gamma^{h*})$ requires:

$$\left| \gamma_1 \cdot \frac{\partial \mathcal{M}}{\partial \Gamma^h} \right| = \gamma_1 \cdot \frac{\theta \mathcal{M}^*(1-\mathcal{M}^*) \cdot \alpha^* B}{1+\alpha^* \Gamma^{w*}} < 1. \tag{26}$$

This condition is satisfied when: (i) the feedback from marriage to human capital (γ_1) is moderate, (ii) the taste shock dispersion (θ) is not too large, or (iii) the marriage rate is not too close to 0.5 (where $\mathcal{M}(1-\mathcal{M})$ is maximized). For the calibrated parameters ($\theta = 3.40$, $\mathcal{M}^* \approx 0.80$, $\alpha^* \approx 0.10$), the left-hand side of (26) equals approximately $0.15\gamma_1$, so stability holds for $\gamma_1 < 6.7$—well above empirically plausible values.

In the presence of this dynamic linkage, Figure 9 illustrates the self-reinforcing decline of fertility, marriage, and gender income gaps.

The decline unfolds through two exogenous drivers and one reinforcing feedback:

(1) **Gender-neutral productivity growth** ($A_t \uparrow$) raises the opportunity cost of children, reducing both single (n^s) and marital (n^m) fertility. Lower n^m reduces men's gains from marriage, lowering $\mathcal{M}$. The resulting composition shift further reduces aggregate fertility n. Freed from childcare, women increase labor supply $l^{♀}$, narrowing the gender income gap Γ^y.

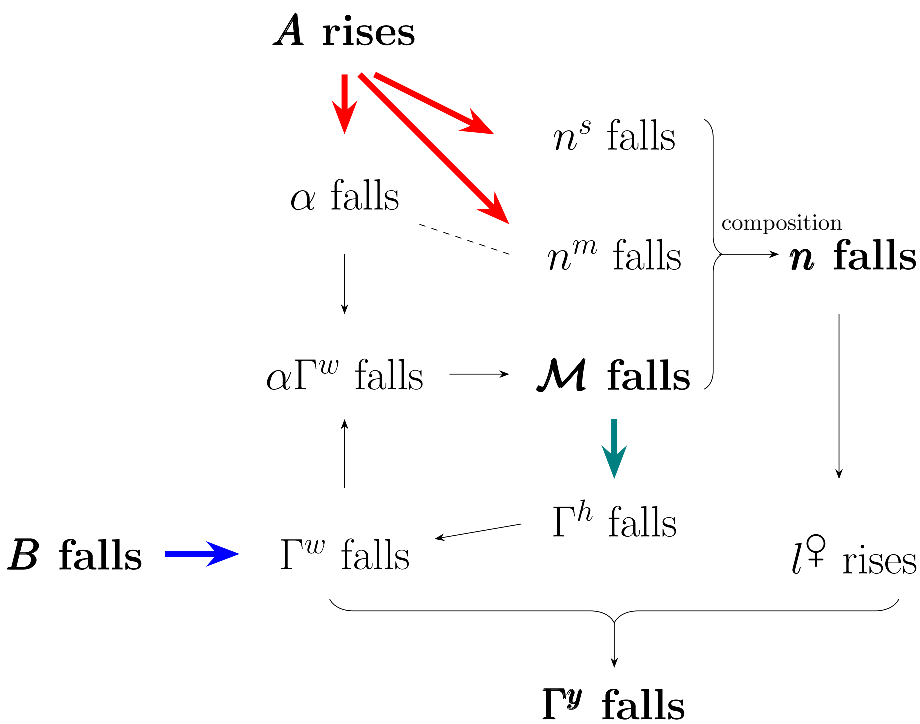


Figure 9. The Dynamic Mechanism

Notes: Red arrows: effects of rising gender-neutral productivity A_t. Blue arrow: effects of falling gender-biased productivity B_t. Teal arrow: dynamic human capital effect via $\mathcal{M}_t \to \Gamma^h_{t+1}$ (Equation (23)).

(2) **Gender-biased productivity decline** ($B_t \downarrow$) narrows the wage gap Γ^w. From women's perspective, lower Γ^w reduces the economic value of marriage ($\alpha\Gamma^w$), lowering $\mathcal{M}$. The composition effect again reduces n, increases $l^{♀}$, and shrinks Γ^y.

(3) **Dynamic propagation via human capital** (feedback loop): Lower $\mathcal{M}_t$ implies that more children grow up without both parents. Since boys suffer greater human capital losses, Γ^h_{t+1} falls (Equation (23)). These less-skilled men become less "marriageable," which further reduces $\mathcal{M}$ and n in the next period and accelerates Γ^y convergence.

Figure 9 offers two key insights. First, both gender-neutral and gender-biased technological changes can simultaneously reduce fertility, marriage rates, and gender income gaps due to their deep interdependence. This perspective complements prior work emphasizing factor-biased technological change, such as Greenwood et al. (2005b) and Ngai and Petrongolo (2017). Second, when family structure affects boys' and girls' human capital formation differently, current shifts in fertility, marriage, and gender gaps may have historical roots. Likewise, today's technological changes can have persistent effects across generations. Thus, policymakers who address any one outcome must look beyond immediate spillovers and account for these long-run feedback effects.

6 Conclusion

This paper establishes a fundamental three-way trade-off: societies rarely achieve high fertility, widespread dual parenthood, and gender income equality simultaneously. Across OECD countries, a global panel of 95 nations, and U.S. states/CZs, the joint attainment of all three outcomes is only a small fraction of what independence would predict—a scarcity that persists across income levels, human capital, and institutional settings.

A unified marriage market model explains this empirical regularity. Fertility and marriage are linked through a two-dimensional price vector—income transfers and marital childbearing—while women's disproportionate childcare burden ties fertility directly to gender income gaps. The result is a structural tension: sustaining any two outcomes requires sacrificing the third. Pro-marriage policies boost dual parenthood but widen income gaps. Gender equality measures narrow gaps but erode marriage and fertility. Policies that reduces overall childcare costs raise fertility at the expense of female labor supply.

A key theoretical contribution concerns the design of childcare policies. Extending the model to allow for gender-differentiated childcare costs reveals that policies redistributing the childcare burden from wives to husbands—such as paternity leave mandates or "use-it-or-lose-it" parental leave provisions—can simultaneously raise fertility, increase marriage rates, and narrow gender income gaps. Two conditions must hold: first, wives' own labor earnings must exceed the transfers they receive from husbands; second, the husband's utility gain from higher fertility must outweigh his consumption loss from increased childcare time. The first condition ensures wives benefit; the second ensures the policy is Pareto-improving within the household.

Quantitatively, the model accounts for Mexico's transformation from 1990 to 2015. Gender-neutral productivity growth explains half the decline in fertility and dual parenthood, while convergence in education and declining labor market bias drive most of the progress toward gender income and welfare equality. Out-of-sample, the calibrated framework accurately predicts Mexico's 2023 fertility and forecasts continued convergence in gender outcomes alongside further erosion of fertility and family structure through 2040.

Extending the model dynamically reveals an important self-reinforcing mechanism. Lower marriage rates disproportionately impair boys' human capital accumulation, rendering future generations of men less "marriageable." This feedback loop amplifies the direct effects of technological progress, accelerating the joint decline in fertility and dual parenthood while hastening gender convergence—even under gender-neutral shocks.

Taken together, these findings deliver a cohesive account of three defining trends of modernity. Fertility, family structure, and gender equality are not independent policy levers;

they are tightly interconnected through women's time allocation, men's marriage incentives, and equilibrium in the marriage market. The analysis identifies a promising policy direction: well-designed policies that promote more equal sharing of childcare responsibilities within marriage may help societies move closer to achieving all three objectives simultaneously—provided that wives' earnings exceed their spousal transfers and husbands value children sufficiently. Where these conditions do not hold, policymakers must recognize the trade-off as a binding constraint and make explicit choices about which objectives to prioritize.

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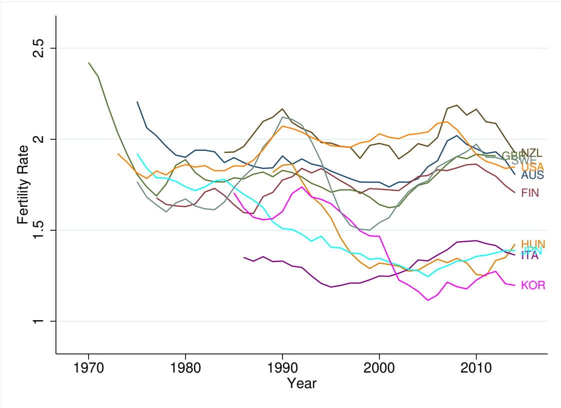
Appendix

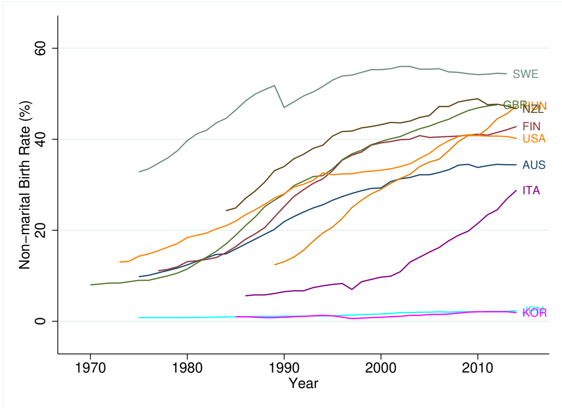
A Additional Empirical Results

A.1 OECD Time-Series Trends

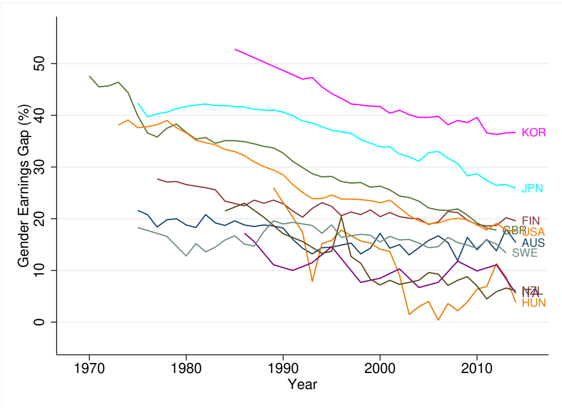
Figure A.1 displays time-series trends in fertility, out-of-wedlock births, and gender earnings gaps for selected OECD countries.



(a) Fertility



(b) Out-of-Wedlock Birth Share



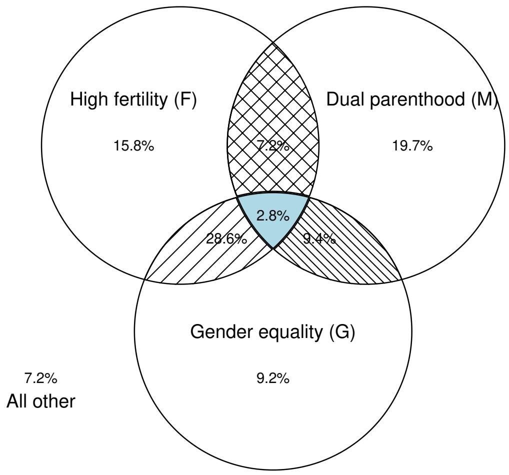
(c) Gender Earnings Gap

Figure A.1. Trends by Country (OECD Sample)

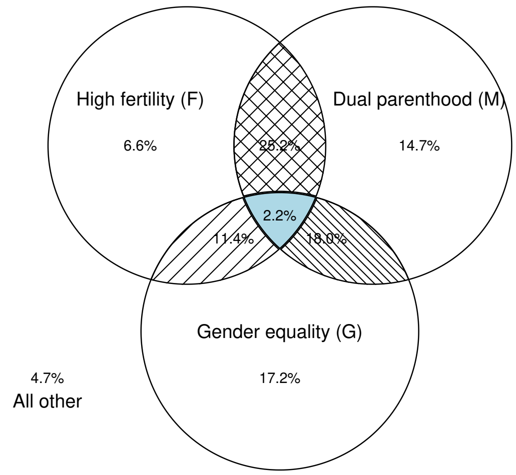
Notes: Time-series trends for OECD countries, 1970–2014. Fertility is the total fertility rate (children per woman). The out-of-wedlock birth share is the percentage of births to unmarried parents. The gender earnings gap is (men's median earnings − women's median earnings) / men's median earnings.

A.2 OECD Subsample Analysis

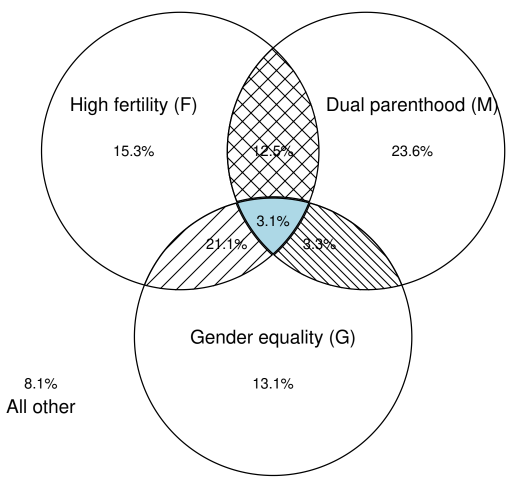
Figure A.2 presents Venn diagrams for OECD subsamples stratified by income and human capital levels.



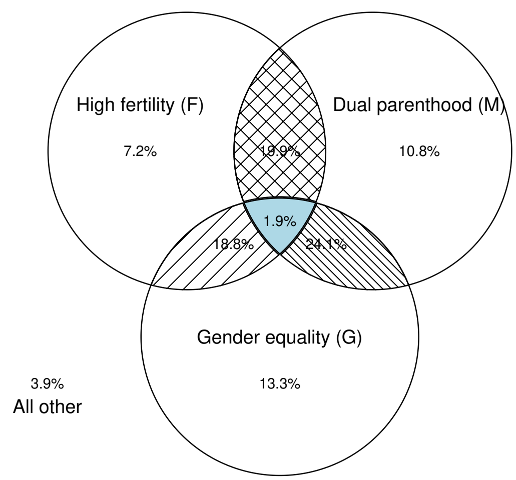
(a) High GDP per Capita



(b) Low GDP per Capita



(c) High Years of Schooling



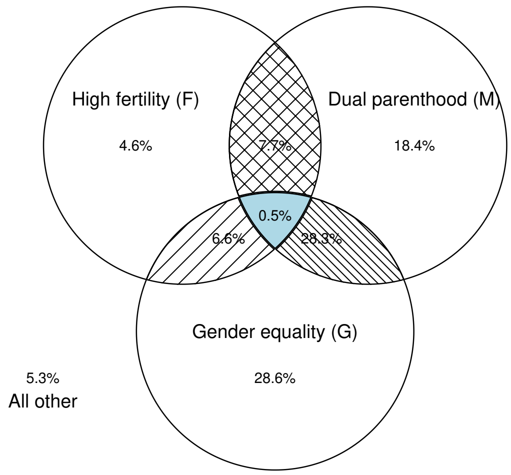
(d) Low Years of Schooling

Figure A.2. Trade-off by Income and Human Capital Levels (OECD Sample)

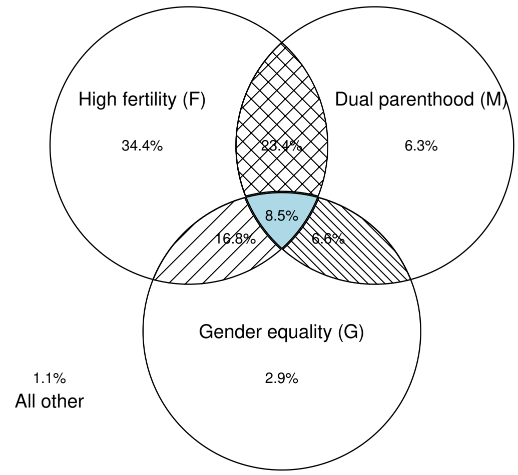
Notes: Venn diagram analysis for subsamples split at median GDP per capita (top row) and median average years of schooling from Barro and Lee (2013) (bottom row). Indicators are redefined at subsample medians in each case. Data is from 37 OECD countries, 1970–2014.

A.3 World Subsample Analysis

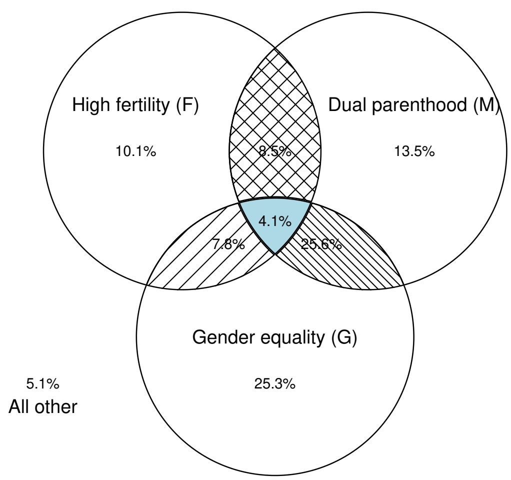
Figure A.3 presents Venn diagrams for World subsamples stratified by income and human capital levels.



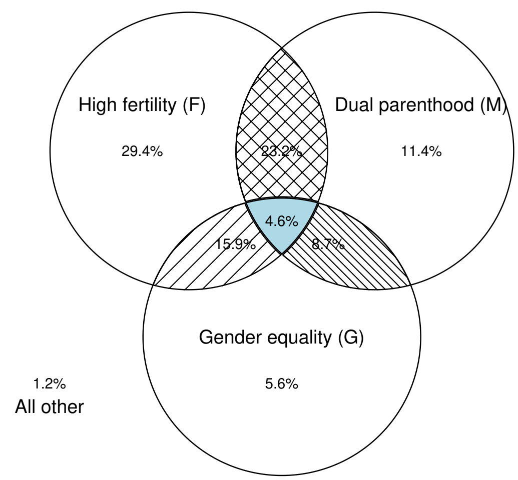
(a) High GDP per Capita



(b) Low GDP per Capita



(c) High Years of Schooling



(d) Low Years of Schooling

Figure A.3. Trade-off by Income and Human Capital Levels (World Sample)

Notes: Venn diagram analysis for subsamples split at median GDP per capita (top row) and median average years of schooling (bottom row). Indicators are redefined at subsample medians in each case. Data is from 95 countries, 1990–2015.

A.4 World Sample Transition Paths

Figure A.4 traces transition paths for all countries in the World sample.

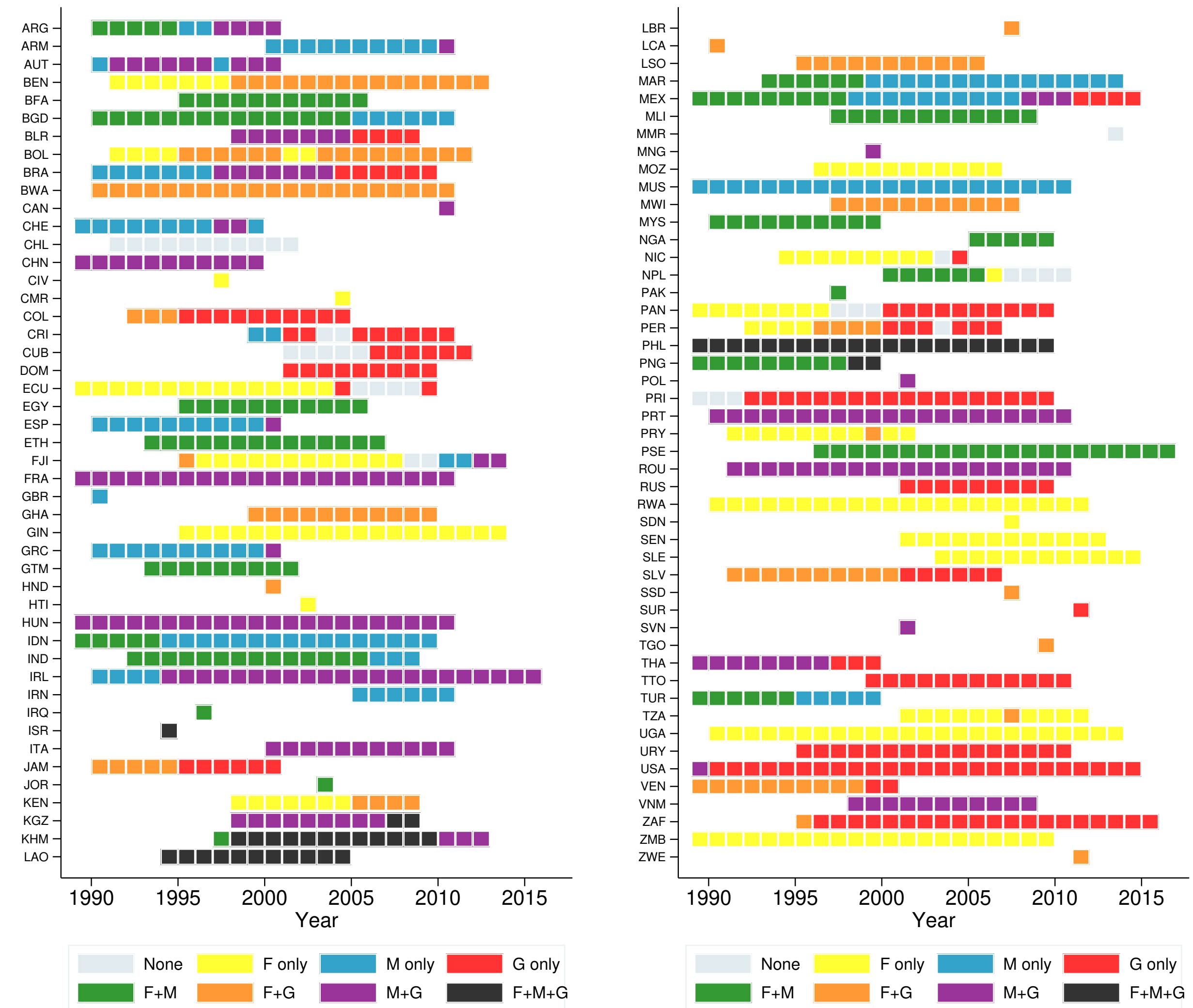


Figure A.4. Transition Paths by Country (World Sample)

Notes: Temporal transitions in the three categories (F = high fertility, M = dual parenthood, G = gender income equality) for all countries in the world sample, 1990–2015.

A.5 U.S. Commuting Zone Analysis

Figure A.5 presents the Venn diagram for U.S. commuting zones.

Figure A.6 maps joint attainment by commuting zone.

A.6 Summary of Mutual Information Results

Table A.1 provides detailed mutual information results across all samples and subsamples.

B Additional Tables and Figures

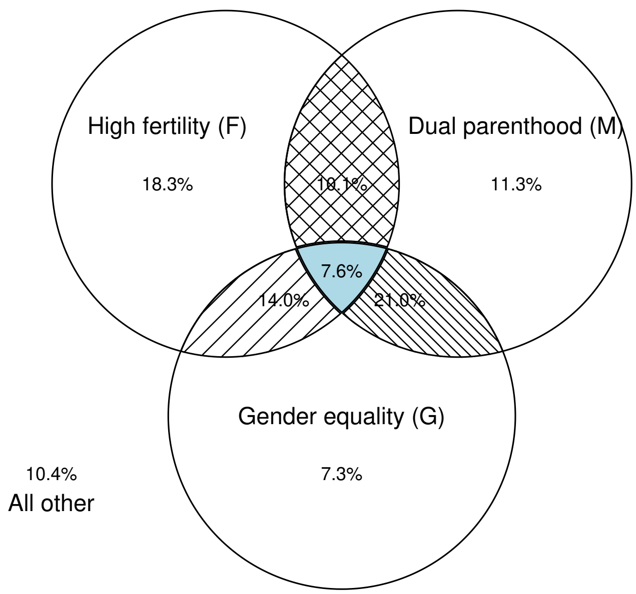


Figure A.5. Trade-off Across U.S. Commuting Zones (2000)

Notes: Venn diagram of high fertility (F: births per 1,000 population > 12.9), dual parenthood (M: single-mother household share < 19.7%), and gender earnings equality (G: gender earnings ratio < 1.80), each defined at cross-CZ medians. $N = 741$ commuting zones.

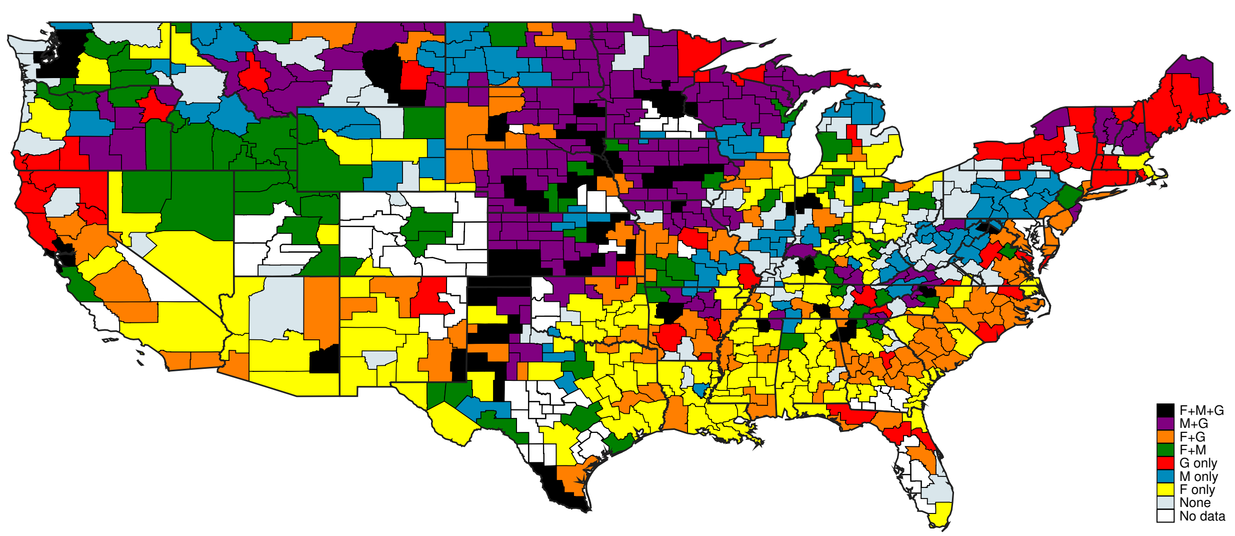


Figure A.6. Joint Attainment by U.S. Commuting Zone (2000)

Notes: Map indicating which outcomes—high fertility (F), dual parenthood (M), gender earnings equality (G)—each commuting zone achieves, based on median thresholds.

Sample	N	Conditional MI $I(F;M\|G)$	Conditional MI $I(F;G\|M)$	Conditional MI $I(M;G\|F)$	Total Corr.	Interaction
OECD						
Full Sample	721	0.078	0.041	0.116	0.165	-0.035^{***}
High Income	360	0.054	0.032	0.129	0.232	+0.017
Low Income	361	0.101	0.059	0.108	0.229	-0.039^{**}
High HC	360	0.049	0.041	0.111	0.209	+0.008
Low HC	361	0.153	0.087	0.113	0.254	-0.099^{***}
World						
Full Sample	1,130	0.061	0.125	0.035	0.169	-0.026^{***}
High Income	587	0.062	0.092	0.034	0.153	-0.035^{***}
Low Income	543	0.034	0.042	0.027	0.096	−0.007
High HC	613	0.039	0.064	0.024	0.113	−0.014
Low HC	517	0.080	0.102	0.050	0.164	-0.032^{**}
United States						
States	51	0.007	0.089	0.029	0.112	−0.007
CZ	741	0.059	0.015	0.017	0.272	+0.001

Table A.1. Detailed Mutual Information Analysis Across Samples

Notes: All values in bits. F = high fertility, M = dual parenthood, G = gender equality. Conditional MI reports $I(X;Y|Z)$, the mutual information between X and Y given Z. Total Corr. = total correlation. Negative interaction information indicates synergy; positive indicates redundancy. Statistical significance based on bootstrap inference with 5,000 replications: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. HC = human capital (years of schooling). CZ = commuting zones.

Country	**Value**	**Year**	**Data Source**
Australia	2.08	2018	Australian Institute of Family Studies
Singapore	2.1	2021	Dollar and Sense (newspaper)
Sweden	2.91	2020	Swedbank's Institute for Personal Finances
Switzerland	3.51	2020	Department of Child Welfare & Career Services, Zurich
Ireland	3.57	2016	Irish Times
Germany	3.64	2018	Federal Statistical Office
United States	4.11	2015	US Dept. of Agriculture
Japan	4.26	2010	Cabinet Office Policy Office of Symbiotic Social
Canada	4.34	2017	Statistics Canada, Money Sense
New Zealand	4.55	2018	Bank of New Zealand Baby Budget Calculator
United Kingdom	5.25	2021	Child Poverty Action Group
Italy	6.28	2021	Consumers Association – Social Promotion Association
China	6.3	2022	China Child-Rearing Cost Report 2024 Edition
South Korea	7.79	2013	Ministry of Health & Welfare

Table A.2. Child-Rearing Cost Relative to Per Capita GDP by Country

Note: This table is adapted from the China Child-Rearing Cost Report 2024 Edition and other national sources as listed. The ratio represents the total cost of raising one child to age 18 divided by per capita GDP in the respective data year.

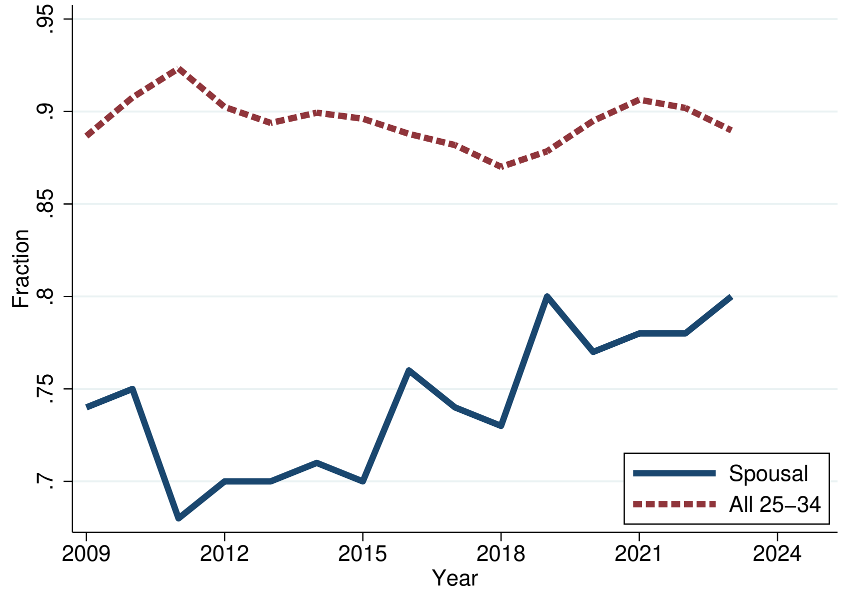
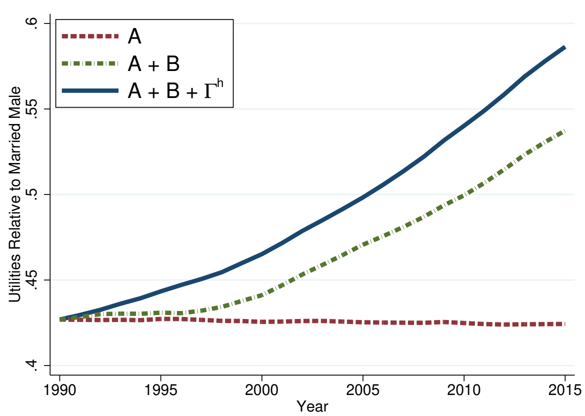
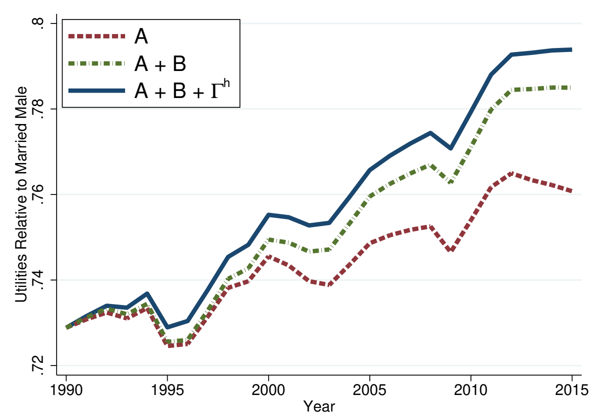


Figure A.7. Gender Earnings Gaps in the U.S.

Notes: The spousal earnings gap is computed using a sample of women from the American Community Survey who are aged 18–39, not enrolled in school, childless, and in their first marriage within the past year to a husband who is also not enrolled in school. The overall gender earnings gap for full-time wage and salary workers aged 25–34 is drawn from the U.S. Bureau of Labor Statistics' Annual Highlights of Women's Earnings.



(a) Relative Utility: $V^{♀,s}/V^{♂,m}$



(b) Relative Utility: $V^{♂,s}/V^{♂,m}$

Figure A.8. Additional Results on Relative Utility

Notes: Solid line: data. Dashed lines: counterfactual paths. "A_t only": vary gender-neutral productivity, fix B_t, Γ_t^h at 1990. "$A_t + B_t$": allow both to vary, fix Γ_t^h. "All": full model.

C Proofs for Baseline Model

C.1 Derivation of Equation (13)

Single men obtain utility $V^{\male,s} = u(w^{\male}, 0) = (1-\beta)^{\frac{\rho}{\rho-1}} w^{\male}$. Married men obtain

$$V^{\male,m} = \Big[(1-\beta)\big((1-\alpha)w^{\male}\big)^{\frac{\rho-1}{\rho}} + \beta (n^m)^{\frac{\rho-1}{\rho}}\Big]^{\frac{\rho}{\rho-1}} \cdot \lambda.$$

Taking the ratio and simplifying:

$$\frac{V^{\male,m}}{V^{\male,s}} = \left[(1-\alpha)^{\frac{\rho-1}{\rho}} + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{\male}}\right)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \cdot \lambda.$$

The marriage market equilibrium condition (7) requires $\frac{V^{\male,m}}{V^{\male,s}} = \left(\frac{\mathcal{M}}{1-\mathcal{M}}\right)^{1/\theta}$. Equating these expressions and solving for α yields equation (13). $\square$

C.2 Proof of Lemma 1

I reduce the two-dimensional fixed-point problem to a single equation in α and exhibit a closed-form solution. To simplify notation, let

$$r \;\equiv\; \frac{\rho-1}{\rho} \;\in\; (0,1), \qquad C \;\equiv\; \left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho} (w^{\female})^{\rho-1} + \chi.$$

For this lemma I replace Assumption 1 by the following condition on primitives, which is strictly weaker in the relevant region of the parameter space and is easily verified in the calibration (see the remark at the end).

Condition (A). The primitives satisfy

$$C \cdot w^{\male} \;>\; \left(\frac{\beta}{1-\beta}\right)^{\rho/(\rho-1)}. \tag{27}$$

Step 1 (Reduction to a single equation in α). The market-clearing condition $\mathcal{M}^{\male} = \mathcal{M}^{\female}$ in (7) is equivalent, by strict monotonicity of the Fréchet c.d.f., to

$$\frac{V^{\male,m}}{V^{\male,s}} = \frac{V^{\female,m}}{V^{\female,s}}.$$

Using the derivation in Appendix C.1 for the left-hand side and Lemma 2 for the right-hand side, this becomes

$$\left[(1-\alpha)^r + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{♂}}\right)^r\right]^{1/r} \cdot \lambda \;=\; \lambda\,(1+\alpha\Gamma^w).$$

Canceling $\lambda > 0$ and raising both sides to the power $r > 0$,

$$(1-\alpha)^r + \frac{\beta}{1-\beta}\left(\frac{n^m}{w^{♂}}\right)^r \;=\; (1+\alpha\Gamma^w)^r. \tag{28}$$

Women's first-order condition (14) pins down n^m as a function of α:

$$n^m(\alpha) \;=\; \frac{1+\alpha\Gamma^w}{C}. \tag{29}$$

Substituting (29) into (28) and factoring out $(1+\alpha\Gamma^w)^r$ on the right,

$$(1-\alpha)^r \;=\; K\cdot(1+\alpha\Gamma^w)^r, \qquad K \;\equiv\; 1 - \frac{\beta}{(1-\beta)\,(Cw^{♂})^r}. \tag{30}$$

Step 2 (Closed-form solution and location). Because $\beta \in (0,1)$, $K < 1$ automatically. Condition (A) is equivalent to $(Cw^{♂})^r > \beta/(1-\beta)$, which is equivalent to $K > 0$. Hence $K \in (0,1)$ and $K^{1/r} \in (0,1)$. Both sides of (30) are positive for any $\alpha \in (-1/\Gamma^w, 1)$, so I may take the positive r-th root:

$$1-\alpha \;=\; K^{1/r}\,(1+\alpha\Gamma^w).$$

Solving for α yields the unique closed form

$$\alpha^* \;=\; \frac{1-K^{1/r}}{1+K^{1/r}\Gamma^w}. \tag{31}$$

Since $K^{1/r} \in (0,1)$, both the numerator and denominator are strictly positive, and the numerator is strictly less than the denominator (because $K^{1/r}\Gamma^w > -K^{1/r}$); hence $\alpha^* \in (0,1)$. The corresponding marital fertility is

$$n^{m*} \;=\; \frac{1+\alpha^*\Gamma^w}{C} \;>\; 0.$$

Step 3 (Uniqueness). Every step above is an equivalence on $(0,1)\times\mathbb{R}_{++}$: the map $x \mapsto x^r$ is a strictly increasing bijection on $\mathbb{R}_+$; (29) is a one-to-one linear relation; and (30) admits the unique solution (31) because the function

$$\phi(\alpha) \;\equiv\; (1-\alpha)^r - K\,(1+\alpha\Gamma^w)^r$$

has derivative

$$\phi'(\alpha) \;=\; -r(1-\alpha)^{r-1} - rK\Gamma^w(1+\alpha\Gamma^w)^{r-1} \;<\; 0 \qquad \text{for all } \alpha \in (0,1),$$

so ϕ is strictly decreasing and can cross zero only once. Therefore any equilibrium $(\alpha, n^m) \in (0,1) \times \mathbb{R}_{++}$ must coincide with (α^*, n^{m*}). $\square$

Remark on Condition (A). Condition (A) is extremely mild. Evaluated at the calibrated parameters ($\beta = 0.104$, $\rho = 1.5$, $\chi = 0.075$), the right-hand side of (27) equals $(0.104/0.896)^3 \approx 1.6\times10^{-3}$, whereas $Cw^{\male}$ exceeds 0.5 throughout the 1990–2015 sample. Condition (A) therefore holds by more than two orders of magnitude. Economically, (27) states that the total cost of raising an additional child (the bracketed term) combined with male labor income is large enough to deliver a strictly positive equilibrium transfer from husbands to wives.

C.3 Proof of Lemma 2

For both married and single women, the first-order condition equating marginal utilities yields the optimal consumption-fertility ratio:

$$\frac{c}{n} = \left(\frac{(1-\beta)w^{\female}\chi}{\beta}\right)^{\rho}.$$

Substituting into the respective budget constraints gives:

$$n^m \cdot \left[\left(\frac{(1-\beta)w^{\female}\chi}{\beta}\right)^{\rho} + w^{\female}\chi\right] = (1+\alpha\Gamma^w)w^{\female}, \tag{32}$$

$$n^s \cdot \left[\left(\frac{(1-\beta)w^{\female}\chi}{\beta}\right)^{\rho} + w^{\female}\chi\right] = w^{\female}. \tag{33}$$

Taking the ratio:

$$\frac{n^m}{n^s} = 1 + \alpha\Gamma^w. \tag{34}$$

Substituting the optimal consumption-fertility ratio into the CES utility function, utility is proportional to fertility:

$$V^{\female,m} = n^m \cdot \Psi \cdot \lambda, \qquad V^{\female,s} = n^s \cdot \Psi,$$

where $\Psi \equiv \left[(1-\beta)\left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho-1}+\beta\right]^{\frac{\rho}{\rho-1}}$ is common to both. Therefore:

$$\frac{V^{♀,m}}{V^{♀,s}} = \lambda \cdot \frac{n^m}{n^s} = \lambda(1+\alpha\Gamma^w). \quad \square$$

C.4 Proof of Proposition 1

By Lemma 2, women's gains from marriage are $V^{♀,m}/V^{♀,s} = \lambda(1+\alpha\Gamma^w)$. An increase in λ directly raises these gains, increasing the marriage rate $\mathcal{M}$ via equation (7).

Since $n^m > n^s$, a higher $\mathcal{M}$ shifts composition toward married women, raising aggregate fertility $n = \mathcal{M}n^m + (1-\mathcal{M})n^s$.

From (18), higher fertility reduces female labor supply: $l^{♀} = 1 - \chi n$ falls. From (17), the gender income gap $\Gamma^y = \Gamma^w / l^{♀}$ therefore widens. $\square$

C.5 Proof of Proposition 2

Step 1: Effect on fertility. From (33), single fertility can be written as:

$$n^s = \frac{1}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho}(w^{♀})^{\rho-1}+\chi}.$$

Since $\rho > 1$, the denominator is strictly increasing in $w^{♀}$, so n^s is strictly decreasing in $w^{♀}$.

From (32), marital fertility is:

$$n^m = \frac{1+\alpha\Gamma^w}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho}(w^{♀})^{\rho-1}+\chi}.$$

When $w^{♀}$ rises, both the numerator falls (since $\Gamma^w = w^{♂}/w^{♀}$ decreases) and the denominator rises. Hence n^m is strictly decreasing in $w^{♀}$.

Step 2: Effect on marriage. Lower n^m reduces men's gains from marriage, lowering the equilibrium transfer α. By Lemma 2, women's gains $\lambda(1+\alpha\Gamma^w)$ fall through both lower α and lower Γ^w, so $\mathcal{M}$ falls.

Step 3: Aggregate effects. With n^s, n^m, and $\mathcal{M}$ all falling, aggregate fertility n unambiguously declines. Female labor supply $l^{♀} = 1 - \chi n$ rises.

Step 4: Gender income gap. The gap is $\Gamma^y = w^{♂}/(w^{♀}l^{♀})$. Since $w^{♀}$ rises and $l^{♀}$ rises, the denominator increases while the numerator is unchanged, so Γ^y falls. $\square$

C.6 Proof of Proposition 3

Step 1: Effect on fertility. From (33):

$$n^s = \frac{1}{\left(\frac{(1-\beta)\chi}{\beta}\right)^\rho (w^{♀})^{\rho-1} + \chi}.$$

Taking the derivative with respect to χ (using the chain rule on χ^ρ in the first term and the additive χ in the second) shows $\partial n^s/\partial\chi < 0$. By analogous reasoning, $\partial n^m/\partial\chi < 0$. Thus both fertilities rise when χ falls.

Step 2: Effect on childcare time. Total childcare time for single women is:

$$T^s = \chi n^s = \frac{1}{\left(\frac{(1-\beta)}{\beta}\right)^\rho \chi^{\rho-1}(w^{♀})^{\rho-1} + 1}.$$

Since $\rho > 1$, I have $\partial T^s/\partial\chi < 0$: when χ falls, T^s rises. Similarly, $T^m = \chi n^m$ rises when χ falls.

Step 3: Effect on marriage. Higher n^m raises men's gains from marriage, increasing the equilibrium transfer α. By Lemma 2, women's gains rise, so $\mathcal{M}$ increases.

Step 4: Gender income gap. Aggregate childcare time χn rises through three channels: (i) χn^s rises, (ii) χn^m rises, and (iii) composition shifts toward marriage where fertility is higher. Hence $l^{♀} = 1 - \chi n$ falls, and $\Gamma^y = \Gamma^w/l^{♀}$ widens. $\square$

D Extended Model with Gender-Differentiated Childcare Costs

D.1 Setup

Within marriage, let $\chi^{♀}$ and $\chi^{♂}$ denote the time costs per child borne by wives and husbands respectively, with $\chi^{♀} > \chi^{♂}$. Single mothers bear the full cost χ. The budget constraints become:

$$c^{♂,m} = (1-\alpha)w^{♂}(1-\chi^{♂}n^m), \tag{35}$$

$$c^{♀,m} = \alpha w^{♂}(1-\chi^{♂}n^m) + w^{♀}(1-\chi^{♀}n^m). \tag{36}$$

Define the wife's *effective income* and *effective childcare cost*:

$$\tilde{w}^{♀} \equiv \alpha w^{♂} + w^{♀}, \qquad \tilde{\chi}^{♀} \equiv \frac{\alpha w^{♂}\chi^{♂} + w^{♀}\chi^{♀}}{\alpha w^{♂} + w^{♀}}.$$

The wife's budget simplifies to $c^{♀,m} = \tilde{w}^{♀}(1 - \tilde{\chi}^{♀} n^m)$.

D.2 Equilibrium Fertility

The first-order conditions yield:

$$n^s = \frac{1}{\left(\frac{(1-\beta)\chi}{\beta}\right)^{\rho} (w^{♀})^{\rho-1} + \chi}, \qquad n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^{♀}}{\beta}\right)^{\rho} (\tilde{w}^{♀})^{\rho-1} + \tilde{\chi}^{♀}}.$$

D.3 Women's Gains from Marriage

Lemma 4. *In the extended model, the wife's utility from marriage and singlehood can be written as:*

$$V^{♀,m} = \lambda \cdot n^m \cdot \Psi^m, \qquad V^{♀,s} = n^s \cdot \Psi^s,$$

where

$$\Psi^m \equiv \left[(1-\beta)\left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho-1} + \beta\right]^{\frac{\rho}{\rho-1}}, \quad \Psi^s \equiv \left[(1-\beta)\left(\frac{(1-\beta)w^{♀}\chi}{\beta}\right)^{\rho-1} + \beta\right]^{\frac{\rho}{\rho-1}}.$$

The wife's gains from marriage are therefore:

$$\frac{V^{♀,m}}{V^{♀,s}} = \lambda \cdot \frac{n^m \cdot \Psi^m}{n^s \cdot \Psi^s}.$$

Proof: The optimal consumption-fertility ratio for married women satisfies $c^{♀,m}/n^m = \left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho}$. Substituting into the CES utility function:

$$\begin{aligned}
u(c^{♀,m}, n^m) &= \left[(1-\beta)(c^{♀,m})^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \\
&= \left[(1-\beta)(n^m)^{\frac{\rho-1}{\rho}}\left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho-1} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \\
&= n^m \cdot \left[(1-\beta)\left(\frac{(1-\beta)\tilde{w}^{♀}\tilde{\chi}^{♀}}{\beta}\right)^{\rho-1} + \beta\right]^{\frac{\rho}{\rho-1}} = n^m \cdot \Psi^m.
\end{aligned}$$

Thus $V^{♀,m} = \lambda \cdot n^m \cdot \Psi^m$. The derivation for single women is analogous. □

D.4 Proof of Proposition 4

Consider a redistribution that increases $\chi^{♂}$ by δ and decreases $\chi^{♀}$ by δ, holding $\chi^{♂} + \chi^{♀}$ constant.

Part (i). Under the redistribution, the wife's effective childcare cost becomes:

$$\tilde{\chi}^{♀}(\delta) = \frac{\alpha w^{♂}(\chi^{♂} + \delta) + w^{♀}(\chi^{♀} - \delta)}{\alpha w^{♂} + w^{♀}}.$$

Differentiating with respect to δ:

$$\frac{\partial \tilde{\chi}^{♀}}{\partial \delta} = \frac{\alpha w^{♂} - w^{♀}}{\alpha w^{♂} + w^{♀}},$$

which is negative if and only if $\alpha w^{♂} < w^{♀}$, i.e., condition (21): $\alpha < w^{♀}/w^{♂}$. $\square$

Part (ii). The wife's effective income $\tilde{w}^{♀} = \alpha w^{♂} + w^{♀}$ is unchanged by the redistribution. From the fertility equation:

$$n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^{♀}}{\beta}\right)^{\rho} (\tilde{w}^{♀})^{\rho-1} + \tilde{\chi}^{♀}},$$

marital fertility depends on δ only through $\tilde{\chi}^{♀}$. The denominator is strictly increasing in $\tilde{\chi}^{♀}$ (both terms increase), so $\partial n^m / \partial \tilde{\chi}^{♀} < 0$. When condition (21) holds, $\partial \tilde{\chi}^{♀} / \partial \delta < 0$, so by the chain rule:

$$\frac{\partial n^m}{\partial \delta} = \frac{\partial n^m}{\partial \tilde{\chi}^{♀}} \cdot \frac{\partial \tilde{\chi}^{♀}}{\partial \delta} = (\text{negative}) \times (\text{negative}) > 0.$$

Single fertility n^s is unchanged since it depends only on χ and $w^{♀}$. $\square$

Part (iii). From Lemma 4, the wife's gains from marriage are $V^{♀,m}/V^{♀,s} = \lambda \cdot (n^m \Psi^m)/(n^s \Psi^s)$. Since n^s and Ψ^s are unaffected by the redistribution, the wife benefits if $\partial(n^m \Psi^m)/\partial \delta > 0$.

To evaluate this, define $A \equiv \left(\frac{1-\beta}{\beta}\right)^{\rho} (\tilde{w}^{♀})^{\rho-1} > 0$, which is constant with respect to $\tilde{\chi}^{♀}$. The fertility equation becomes $n^m = 1/D$ where $D = A(\tilde{\chi}^{♀})^{\rho} + \tilde{\chi}^{♀}$.

From the proof of Lemma 4, I can show that:

$$\Psi^m = \left[\beta \left(A(\tilde{\chi}^{♀})^{\rho-1} + 1\right)\right]^{\frac{\rho}{\rho-1}}.$$

Noting that $D = \tilde{\chi}^{♀}(A(\tilde{\chi}^{♀})^{\rho-1}+1)$, I have $A(\tilde{\chi}^{♀})^{\rho-1}+1 = D/\tilde{\chi}^{♀}$, so:

$$\Psi^m = \beta^{\frac{\rho}{\rho-1}} \left(\frac{D}{\tilde{\chi}^{♀}}\right)^{\frac{\rho}{\rho-1}}.$$

The product $n^m\Psi^m$ simplifies to:

$$n^m\Psi^m = \frac{1}{D} \cdot \beta^{\frac{\rho}{\rho-1}} \left(\frac{D}{\tilde{\chi}^{♀}}\right)^{\frac{\rho}{\rho-1}} = \beta^{\frac{\rho}{\rho-1}} \cdot D^{\frac{1}{\rho-1}} \cdot (\tilde{\chi}^{♀})^{-\frac{\rho}{\rho-1}}.$$

Define $\xi \equiv A(\tilde{\chi}^{♀})^{\rho-1} > 0$. Then $D = \tilde{\chi}^{♀}(\xi+1)$, so:

$$n^m\Psi^m = \beta^{\frac{\rho}{\rho-1}} \cdot (\tilde{\chi}^{♀})^{\frac{1}{\rho-1}}(\xi+1)^{\frac{1}{\rho-1}} \cdot (\tilde{\chi}^{♀})^{-\frac{\rho}{\rho-1}} = \beta^{\frac{\rho}{\rho-1}} \cdot (\tilde{\chi}^{♀})^{-1} \cdot (\xi+1)^{\frac{1}{\rho-1}}.$$

Differentiating with respect to $\tilde{\chi}^{♀}$, using $\frac{\partial \xi}{\partial \tilde{\chi}^{♀}} = \frac{(\rho-1)\xi}{\tilde{\chi}^{♀}}$:

$$\begin{aligned}
\frac{\partial(n^m\Psi^m)}{\partial \tilde{\chi}^{♀}} &= \beta^{\frac{\rho}{\rho-1}} \left[-(\tilde{\chi}^{♀})^{-2}(\xi+1)^{\frac{1}{\rho-1}} + (\tilde{\chi}^{♀})^{-1} \cdot \frac{1}{\rho-1}(\xi+1)^{\frac{1}{\rho-1}-1} \cdot \frac{(\rho-1)\xi}{\tilde{\chi}^{♀}}\right] \\
&= \beta^{\frac{\rho}{\rho-1}} \left[-\frac{(\xi+1)^{\frac{1}{\rho-1}}}{(\tilde{\chi}^{♀})^2} + \frac{\xi(\xi+1)^{\frac{2-\rho}{\rho-1}}}{(\tilde{\chi}^{♀})^2}\right] \\
&= \frac{\beta^{\frac{\rho}{\rho-1}}(\xi+1)^{\frac{2-\rho}{\rho-1}}}{(\tilde{\chi}^{♀})^2}\left[\xi - (\xi+1)\right] \\
&= -\frac{\beta^{\frac{\rho}{\rho-1}}(\xi+1)^{\frac{2-\rho}{\rho-1}}}{(\tilde{\chi}^{♀})^2} < 0.
\end{aligned}$$

The inequality holds because all factors in the final expression are positive for $\rho > 1$. Therefore, $\partial(n^m\Psi^m)/\partial\tilde{\chi}^{♀} < 0$ unconditionally.

When condition (21) holds, $\partial\tilde{\chi}^{♀}/\partial\delta < 0$, so:

$$\frac{\partial(n^m\Psi^m)}{\partial\delta} = \frac{\partial(n^m\Psi^m)}{\partial\tilde{\chi}^{♀}} \cdot \frac{\partial\tilde{\chi}^{♀}}{\partial\delta} = (\text{negative}) \times (\text{negative}) > 0.$$

The wife always benefits from redistribution when condition (21) holds. No additional restriction on ρ is required. □

Part (iv). The husband's utility from marriage is:

$$V^{♂,m} = \lambda \cdot u(c^{♂,m}, n^m) = \lambda\left[(1-\beta)(c^{♂,m})^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}},$$

where $c^{♂,m} = (1-\alpha)w^{♂}(1-\chi^{♂}n^m)$.

Differentiating $V^{\male,m}$ with respect to δ:

$$\frac{dV^{\male,m}}{d\delta} = \lambda \cdot \frac{\rho}{\rho-1} \cdot U^{\frac{1}{\rho-1}} \left[(1-\beta)(c^{\male,m})^{-\frac{1}{\rho}} \frac{dc^{\male,m}}{d\delta} + \beta(n^m)^{-\frac{1}{\rho}} \frac{dn^m}{d\delta}\right],$$

where $U \equiv (1-\beta)(c^{\male,m})^{\frac{\rho-1}{\rho}} + \beta(n^m)^{\frac{\rho-1}{\rho}} > 0$.

Since $\lambda > 0$, $\frac{\rho}{\rho-1} > 0$ (for $\rho > 1$), and $U^{\frac{1}{\rho-1}} > 0$, the sign of $\frac{dV^{\male,m}}{d\delta}$ equals the sign of the bracketed term.

Computing $\frac{dc^{\male,m}}{d\delta}$: From the husband's budget constraint:

$$c^{\male,m} = (1-\alpha)w^{\male}(1-\chi^{\male} n^m).$$

Taking the total derivative with respect to δ:

$$\frac{dc^{\male,m}}{d\delta} = (1-\alpha)w^{\male}\left[-n^m \frac{d\chi^{\male}}{d\delta} - \chi^{\male}\frac{dn^m}{d\delta}\right] = (1-\alpha)w^{\male}\left[-n^m - \chi^{\male}\frac{dn^m}{d\delta}\right],$$

where I used $\frac{d\chi^{\male}}{d\delta} = 1$ (redistribution increases the husband's childcare cost one-for-one with δ).

Since $n^m > 0$, $\chi^{\male} \geq 0$, and $\frac{dn^m}{d\delta} > 0$ (from Part ii when condition (21) holds), I have $\frac{dc^{\male,m}}{d\delta} < 0$: the husband's consumption falls.

Condition for husband to benefit: The husband benefits ($\frac{dV^{\male,m}}{d\delta} > 0$) if and only if:

$$\beta(n^m)^{-\frac{1}{\rho}}\frac{dn^m}{d\delta} > -(1-\beta)(c^{\male,m})^{-\frac{1}{\rho}}\frac{dc^{\male,m}}{d\delta}.$$

Substituting the expression for $\frac{dc^{\male,m}}{d\delta}$:

$$\beta(n^m)^{-\frac{1}{\rho}}\frac{dn^m}{d\delta} > (1-\beta)(c^{\male,m})^{-\frac{1}{\rho}}(1-\alpha)w^{\male}\left[n^m + \chi^{\male}\frac{dn^m}{d\delta}\right].$$

Rearranging:

$$\frac{\beta}{1-\beta} \cdot \left(\frac{c^{\male,m}}{n^m}\right)^{\frac{1}{\rho}} \cdot \frac{dn^m}{d\delta} > (1-\alpha)w^{\male}\left[n^m + \chi^{\male}\frac{dn^m}{d\delta}\right].$$

Dividing both sides by $\frac{dn^m}{d\delta} > 0$:

$$\frac{\beta}{1-\beta} \cdot \left(\frac{c^{\male,m}}{n^m}\right)^{\frac{1}{\rho}} > (1-\alpha)w^{\male}\left[\frac{n^m}{dn^m/d\delta} + \chi^{\male}\right],$$

which is condition (22). $\square$

Comparative statics on condition (22):

- *Effect of* β: The left-hand side is increasing in β (both through $\frac{\beta}{1-\beta}$ and indirectly through higher n^m). A higher preference weight on children makes the husband more likely to benefit.

- *Effect of* $\chi^{\male}$: The right-hand side is increasing in $\chi^{\male}$. When the husband already bears a substantial childcare burden, the marginal consumption loss from additional childcare is larger, making him less likely to benefit.

- *Effect of* ρ: The left-hand side involves $(c^{\male,m}/n^m)^{1/\rho}$. Since typically $c^{\male,m} > n^m$ in units, a lower ρ (closer to 1) raises the left-hand side, making the husband more likely to benefit. Intuitively, when consumption and children are less substitutable, the marginal utility of additional children is higher.

- *Effect of fertility response*: A larger $\frac{dn^m}{d\delta}$ reduces $\frac{n^m}{dn^m/d\delta}$ on the right-hand side, making the condition easier to satisfy. The fertility response is larger when condition (21) holds with greater slack (i.e., $\alpha \ll w^{\female}/w^{\male}$).

Parts (v)–(vi). When both spouses benefit from redistribution, each spouse's utility ratio $V^{g,m}/V^{g,s}$ increases. From the marriage market equilibrium condition (7):

$$\mathcal{M}^g = \frac{1}{1 + (V^{g,s}/V^{g,m})^{\theta}},$$

higher values of $V^{g,m}/V^{g,s}$ for both genders imply a higher equilibrium marriage rate $\mathcal{M}$.

Aggregate fertility is:

$$n = \mathcal{M} \cdot n^m + (1-\mathcal{M}) \cdot n^s.$$

I have established that: (i) n^m increases (Part ii); (ii) n^s is unchanged; (iii) $\mathcal{M}$ increases when both spouses benefit (Part v); and (iv) $n^m > n^s$ in equilibrium (married women have higher fertility due to income transfers). All effects work in the same direction, so aggregate fertility n unambiguously increases. $\square$

E Proof for Dynamic Extension

E.1 Proof of Lemma 3

Let $G \equiv \lambda(1 + \alpha\Gamma^w)$ denote women's gains from marriage. The marriage rate is:

$$\mathcal{M} = \frac{G^\theta}{1 + G^\theta}.$$

Differentiating:

$$\frac{\partial \mathcal{M}}{\partial G} = \frac{\theta G^{\theta-1}}{(1 + G^\theta)^2} = \frac{\theta}{G} \cdot \frac{G^\theta}{1 + G^\theta} \cdot \frac{1}{1 + G^\theta} = \frac{\theta}{G} \cdot \mathcal{M}(1 - \mathcal{M}).$$

Since $\Gamma^w = B \cdot \Gamma^h$, I have $\frac{\partial G}{\partial \Gamma^h} = \lambda\alpha B$. By the chain rule:

$$\frac{\partial \mathcal{M}}{\partial \Gamma^h} = \frac{\theta}{G} \cdot \mathcal{M}(1 - \mathcal{M}) \cdot \lambda\alpha B = \frac{\theta\mathcal{M}(1 - \mathcal{M}) \cdot \alpha B}{1 + \alpha\Gamma^w} > 0. \quad \square$$